The Geometric Engine of Time: The Fokker-Planck Prime Sieve in Quaternionic Lattices

BioChain AI — SHD-CCP Architecture Research

Consolidated Technical Report (Continued Series)

Abstract

This installment of The Geometric Engine of Time formalizes the Fokker-Planck prime sieve on quaternionic lattices and proves what it can and cannot do. On a lattice the sieve is a master equation whose Gibbs state is positive on every node; absorbing prime traps send all mass to the primes with the harmonic measure, and on the infinite Hurwitz lattice the sieve is defined exactly when the Hurwitz partition function converges. Drift along lattice lines without a potential cannot sieve: by the cycle-current formula of the cubit installment it carries a current and produces entropy, yet its stationary state is uniform. The sieve therefore does not compute the primes. Under the series' Dual-Helix principle it proposes trap sites and exact arithmetic certifies them, and the Engine's 30-node wheel gives the proposer a proven bias of 3.75, the largest of any primorial wheel whose unit classes fit the packet's 3-bit wheel field. The sieve lives in four dimensions. The golden Clifford torus contains no lattice point, while its Fibonacci approximants carry the points (F_m, F_{m+1}, F_{m−1}, F_m), whose prime norms beyond 3, namely 7, 47 and 2207, are congruent to 7 mod 8 and so missing from every three-dimensional lattice. In exact arithmetic the sieve has no drift, and a periodic delta schedule returns to the identity within 12 repetitions or never. The quaternions 1 ± 2i, 1 ± 2j, 1 ± 2k turn the six steps of the cubic lattice into a free group whose words form a 6-regular hyperbolic tree over the cubic lattice; every address factors back to its unique word, and reduction modulo a prime gives Ramanujan graphs on which the sieve's diffusion mixes in 3.25 time units against 20.15 on a cubic torus of the same size. On the Einstein Tile the sieve reads the Form 1 Hurwitz core in exact 64-bit arithmetic and certifies prime-norm anchor packets with a 2,017-byte table. Anchored to the realized cubit, whose uncertainty caps the informative quadrance near 2.3 × 10^7, the Triple Quad Formula stays exact in 64-bit integers, where double precision declares a non-collinear lattice triangle collinear. Together these define a universal lattice standard on which agents with no shared orientation, medium or history align non-locally by invariants and locally by Horn's quaternion method, with every alignment certified exactly and recorded under a Merkle root. Its use for AI communication, alignment and governance is stated as a design proposal; the mathematics beneath it is proven, established or computed.

Editorial Note. This document is an installment of *The Geometric Engine of Time* (Hoard 2026a), cited in the series as Hoard (2026e), and formalizes the source note *The Fokker-Planck Prime Sieve in Quaternionic Lattices*. It contains the sieve, its proofs and what it enables. The status of every claim, the audit of the source note and of the companion papers, the open problems with their test protocols, and the relation to earlier versions are collected in its companion, *Status of Claims, Audit and Open Problems* (Hoard 2026h). The paper imports the companion papers of the SHD-CCP research programme, cited as BioChain AI (2026a–i), the proofs of *The Fifteen Lemmas of the Geometric Engine* (Hoard 2026, cited as Fifteen Lemmas), and the metrology and exact results of *A Proposal for a Standardization of the Cubit* (Hoard 2026d), whose conventions it follows. All quantitative claims are annotated with status tags: **[PROVEN]**, **[HEURISTIC]**, **[CONJECTURAL]**, **[OPEN]**, **[SUPERSEDED]**, refined by five tags defined under Epistemic Conventions and Scope.

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Epistemic Conventions and Scope

Every claim carries one of ten tags: the nine of the series, as fixed in the cubit installment (Hoard 2026d, Table 1), and **[CORRESPONDENCE]**, adopted from the Fifteen Lemmas. Claims taken from a companion paper keep the tag that paper's own argument supports, which is not always the tag the paper gave itself.

Table 1: Status tags used in this installment.

Tag	Meaning	Example in this paper
[PROVEN]	A deductive result with a complete proof in this document. Series tag; exact arithmetic on stated inputs is tagged **[COMPUTED]**.	No lattice point on the golden torus (Theorem 5.2)
[CORRESPONDENCE]	Two finite sets of equal size, or two equal numbers; no structure-preserving map is claimed. Adopted from the Fifteen Lemmas.	$W_\Sigma = 60$ equals the exponent of 2I (Section 6)
[DEFINITION]	A convention adopted here. It can be useful or not, but not true or false.	The sieve's reading of the Einstein Tile (Definition 8.1)
[ESTABLISHED]	Standard mathematics, cited and not re-proved.	Lubotzky–Phillips–Sarnak Ramanujan graphs
[COMPUTED]	A number obtained by evaluating a stated formula. Reproducible from Appendix A.	Mixing time 3.25 on X(5, 13)
[EMPIRICAL]	A comparison with measured data.	Not used in this paper
[HEURISTIC]	A modelling choice motivated by analogy, not derived. Series tag.	The lattice standard as an AI alignment layer
[CONJECTURAL]	A claim that would need new evidence to be accepted. Series tag.	Oracle bias above 1 for factoring (BioChain AI 2026a)
[OPEN]	A stated question awaiting proof or test. Series tag.	A lattice line with infinitely many prime-norm nodes (companion, Problem 2)
[SUPERSEDED]	A claim of a source replaced by a corrected statement. Series tag.	The source note's prime-basis expansion (Section 2.1)

Rule for mixed claims. As in the cubit installment (Hoard 2026d), a computed value of a heuristic formula is tagged **[HEURISTIC / COMPUTED]**: the arithmetic is checked, but the modelling choice is not thereby established. The precision horizons of Section 8.2 are the example here.

What this paper does not claim.

- It does not claim a faster primality test, prime sieve or factoring method. Every construction that places a prime trap uses an exact primality test.
- It does not claim that a geometric standard solves AI alignment. Section 9 proposes a communication and certification layer, tagged **[HEURISTIC]** where it goes beyond the mathematics.
- It does not claim that the quaternionic lattice is a physical system. The Fokker-Planck dynamics are a mathematical model; the cubit gives lattice addresses a physical scale (Section 8.1), not the dynamics a physical mechanism.

Relation to the series. Section 1 maps every source, every lemma of the Fifteen Lemmas and every result of the cubit installment used here to the sections that use it. Section 10 lists where the Geometric Engine of Time supplies the most accurate method of calculation, and where it is only the source of a construction.

Relation to the companion. This paper states and proves. Where it replaces a claim of the source note or of a companion paper, it states the replacement and leaves the comparison to the companion (Hoard 2026h), which also holds the consolidated status of claims, the errata, the open problems and the history of this installment. Appendix A verifies the numbers of both documents.

Notation. Inline mathematics is set in plain text (for example $N(q) = a^2 + b^2 + c^2 + d^2$); display equations are typeset.

1 Sources

The paper formalizes one source note and draws on fifteen further sources: four installments of the Engine series, the Fifteen Lemmas monograph, nine companion papers of the SHD-CCP programme and one external paper on quantum codes.

1.1 The Sources and Where They Enter

Each row of Table 2 names a source, what it supplies and the sections that use it. Section 10 states where a series construction is the most accurate way to calculate and where it is a source only.

Table 2: Where each source enters this paper.

Label	Paper	What it supplies	Sections
Source note	*The Fokker-Planck Prime Sieve in Quaternionic Lattices*	The master equation, the prime potential and traps, TQF lines, the precision horizon	2, 3, 4, 8
Hoard 2026a	*The Geometric Engine of Time: Mathematical Architecture of the 6-Channel, 5-Tier Toroidal Orbital Calculator*	$W_\Sigma = 60$, the 30-node closure, the TQF area invariant, the $C_3 \times C_2$ channels, the alternating embedding	3, 4, 6, 7.5, 8, 9, 10

Label	Paper	What it supplies	Sections
Hoard 2026b	*The Torsional Markov Memory-Offload Pump*	Cycle currents of a non-gradient drift	2.3
Hoard 2026c	*The Hyperbolic Sparsemax Pump*	The rational Lorentz lift and the kernel Q_h	7.6
Hoard 2026d	*A Proposal for a Standardization of the Cubit*	Tag conventions; the cycle-current formula; the base-6 and $\mathbb{Z}/30$ wheels; the realized cubit, its uncertainty and medium-free modes; the joint 0.5 s closure; the cube group of the tier rotations	2.3, 3.1, 4.2, 8, 9, 10
Fifteen Lemmas	*The Fifteen Lemmas of the Geometric Engine* (Hoard 2026)	Proofs of Lemmas 1 to 15 and the Einstein Tile	2 to 10
BioChain AI 2026a	*A Unified Dual-Helix Framework for Geometrically-Guided Pollard's Rho*	Geometry proposes, arithmetic certifies; the FP4 sampler; the Lorentz lift; the Oracle Bias Hypothesis	2, 3, 7.6, 9
BioChain AI 2026b	*The Golden Strassen-Clifford Geodesic Embedding*	The Helix A and Helix B roles	3
BioChain AI 2026c	*Quaternionic Delta Encoding on Golden Strassen-Clifford Toroids*	The golden Clifford torus and its delta encoding	5
BioChain AI 2026d	*Topological Symmetry Folding and Hyperbolic Projection in the SHD-CCP Framework*	The hyperbolic projection; the question whether it runs on shifts	6, 7
BioChain AI 2026e	*SHD-CCP Review: The 3-Nested Decoding Framework*	The translation requirement, the S-plane scan with TQF pruning, Codex crystallization	8, 9
BioChain AI 2026f	*The Chiral Compression Engine*	Dual-Helix governance and Proof-of-Manifold	9.6
BioChain AI 2026g	*SHD-CCP Review: The Trefoil Projection Protocol*	16 fibres × 4 phases as an addressing of the 64 bits	8.1
BioChain AI 2026h	*Quaternion Equilibrium and Voxel Alignment*	S^3 parallelizability, the scalar key, the master quaternion, the coherence Φ	4.1, 7.6, 9.3
BioChain AI 2026i	*Proof: Triple-Quad Function on a 3-Folded Packet*	Scale-normalized quads	9.3
Butt et al. 2026	*Complementary 3D color codes for transversal quantum logic* (arXiv:2607.05107)	Complementary encodings, a one-way transversal CNOT, invariant stabilizers	9.5

Editorial Note. Two labelling points. The series cites this installment as Hoard (2026e), the Fifteen Lemmas audit and protocol notes as Hoard (2026f) and (2026g), and the companion to this paper as Hoard (2026h), so the companion papers of the SHD-CCP programme are cited as BioChain AI (2026a–i), as the Fifteen Lemmas cite them. One source is not a series paper: *Complementary 3D color codes for transversal quantum logic* is by F. Butt, L. Colmenarez, E. Weilandt, T. Peham, R. Wille and M. Müller (RWTH Aachen, Forschungszentrum Jülich and TU Munich), and is cited as external work.

1.2 The Fifteen Lemmas in This Paper

Nine of the fifteen lemmas carry proofs this paper uses. Lemma 3 enters only through Lemma 5, Lemma 12 is used in the companion, and Lemmas 1, 2, 6 and 10 do not bear on the sieve. Every imported statement was re-checked before use (Appendix A, sections W to Y).

Table 3: Results of the Fifteen Lemmas used in this paper.

Fifteen Lemmas result	Used in	What it does here
Lemma 4, Parts 2, 4 and 5	3.1, 8.2, 10	Weitzenböck bound $A_R \le S^2/3$; the powers of the tier spinor h; the Legendre split of the wheel
Lemma 5, Parts 2, 4, 5 and 6(a)	4.1, 8.2, 8.3, 9.4	The cube group O and its lift 2O; the primes of the box $[-2, 2]^4$; a lattice proof of Theorem 8.4; the Gram form of the TQF
Lemma 7, Part 3(d) and (e)	2.3, 9.6	Drift without a potential does not sieve; four records closed by a product survive one loss
Lemma 8, Step 2 and Propositions 6.2 to 6.5	4.1, 5.1, 7.4, 7.6	The matrix L(q); the ghost map; the inclusion $SU(2) \subset SL(2, \mathbb{C})$; the {4,3,5} honeycomb; the trefoil on the golden torus
Lemma 9, Parts 1 to 3	7.7, 9.4	Triplet classes mod 30; CRT addresses with closed-form inverses; the rational Cayley map
Lemma 11, Parts 2, 3 and 5; Proposition 11.4	5.1, 5.3, 9.2	Density depends on frequencies, not radii; the two-gap golden circle; the 30-pulse clock window
Lemma 13, Parts 1 to 3	6, 7.1	Rational finite orders are 1, 2, 3, 4 or 6; denominators grow without bound; integer bit growth
Lemma 14, Parts 1 to 4	2, 3.1	The Gibbs and harmonic measures; the infinite Hurwitz lattice; the Circularity Boundary
Lemma 15	8.2, 9.6	Merkle roots certify; an 8-bit header only filters
§6.1, the Einstein Tile	8	The packet on which the sieve runs
§7.2 and Theorems 7.5, 7.8, 7.10 and 7.19	7.1, 7.6, 9.3, 9.6	History loss in unit chains; the hidden spin sign; what an exchange preserves; generic freeness

1.3 The Cubit Standard in This Paper

The cubit installment (Hoard 2026d) contributes a physical unit and five exact results. Each strengthens a step of the sieve or of the standard built on it, and each was re-checked in Appendix A, section Z.

Table 4: Results of the cubit installment used in this paper.

Cubit result (Hoard 2026d)	Used in	What it does here
Proposition 4.8, the cycle current, with Kolmogorov's criterion and entropy production	2.3	Prices the drift without a potential: a positive current and positive entropy production, with no selection
Theorem 4.9 and the Remark on the 30-node wheel	3.1	Base-6 against $\mathbb{Z}/30$; $\mathbb{Z}/30$ is the best wheel that fits the tile's 3-bit field (Proposition 3.3)
Section 3.5, the uncertainty budget, and Remark 4.3, numerical conditioning	8.2	A physical precision horizon near $Q \approx 2.3 \times 10^7$; why the TQF is evaluated in integers
Section 3.2, Proposition 3.1 and Lemma 3.2	8.1, 9.2, 9.3	The cubit as the unit of the scale register; the $4 \times 4 \times 4$ voxel cube of edge $\lambda/4$; medium-free mode invariants
Section 4.1, shortest and joint closures	9.2, 10	The 0.5 s window shared by caesium, 40 Hz and 432 Hz, proved there first
Proposition 4.6(3) and Appendix E	4.2, 9.4, 10	Stated sign conventions for rotations; the Engine's tier rotations generate the cube group O

2 The Lattice Fokker-Planck Equation

On a lattice the Fokker-Planck equation is a master equation; its stationary state is positive on every node, so exact support on primes needs absorbing traps, and the Dual-Helix sampler of the series behaves the same way.

Status: [ESTABLISHED] and [PROVEN]. Sources: source note §§1–2; BioChain AI (2026a), §3.7, equation (23), read here as a killing term; Hoard (2026b) for non-gradient drift, made exact by Fifteen Lemmas, Lemma 7 Part 3(e), and priced by the cycle-current formula of Hoard (2026d), Proposition 4.8; Fifteen Lemmas, Lemma 14, which proves Propositions 2.2 and 2.3 independently and extends them to the infinite Hurwitz lattice.

2.1 The Master Equation and Its Stationary State

The source note writes the continuum equation:

$$\frac{\partial \rho}{\partial t} = -\sum_i \frac{\partial}{\partial q_i}\left[D_1^{(i)}(\mathbf{q})\,\rho\right] + \sum_{i,j} \frac{\partial^2}{\partial q_i\,\partial q_j}\left[D_2^{(i,j)}(\mathbf{q})\,\rho\right]$$

Derivatives in q are not defined on lattice nodes, so the evolution of a probability vector on a lattice is a master equation, the generator of a continuous-time Markov chain.

Definition 2.1 (Lattice Master Equation). [DEFINITION] Formalizes source §1.

Let Λ be a finite set of lattice nodes joined by nearest-neighbour edges, V a real potential on Λ and D > 0 a diffusion constant. The probability vector ρ(t) evolves by:

$$\frac{d\rho_b}{dt} = \sum_{a\sim b}\left(\rho_a\,k_{ab} - \rho_b\,k_{ba}\right), \qquad k_{ab} = D\,e^{-(V_b - V_a)/2}$$

Its continuum limit is a Fokker-Planck equation with drift proportional to −∇V and constant diffusion. The note's prime potential, with minima only where the quadrance Q is prime, depends on a node only through Q; its gradient is radial, not along TQF lines, and placing its minima requires knowing which quadrances are prime **[HEURISTIC]**.

Proposition 2.2 (Gibbs Stationary State). [PROVEN] Formalizes source §§1–2; also Fifteen Lemmas, Lemma 14 Part 1.

For the master equation of Definition 2.1 on a connected finite lattice, the unique stationary distribution is:

$$\rho_a^{\mathrm{ss}} = \frac{e^{-V_a}}{\sum_{b\in\Lambda} e^{-V_b}}$$

It is strictly positive on every node, so no stationary state with positive diffusion is supported only on prime-quadrance nodes.

Proof. Since e^(−V_a) k_ab = D e^(−(V_a + V_b)/2) = e^(−V_b) k_ba, the Gibbs vector satisfies detailed balance on every edge and is stationary. A connected finite chain is irreducible, so its stationary distribution is unique and strictly positive (Perron–Frobenius). ■

Remark (Concentration Against Speed). [COMPUTED] On the 4 × 4 × 4 lattice with coordinates 0 to 3, set V = 0 on the 25 prime-quadrance nodes and V = ΔV on the other 39,

with D = 1. Raising the barrier concentrates mass on prime nodes but slows relaxation: the spectral gap falls 29-fold between ΔV = 0 and ΔV = 8 (Table 5).

Table 5: Gibbs mass on the prime-quadrance nodes of the 4 × 4 × 4 lattice.

Barrier ΔV	Prime-node mass	Spectral gap (units of D)	Smallest node weight
0	0.3906	0.5858	1.56×10^{-2}
1	0.6354	0.5371	9.35×10^{-3}
2	0.8257	0.3984	4.47×10^{-3}
4	0.9722	0.1548	7.12×10^{-4}
8	0.9995	0.0203	1.34×10^{-5}

The note's "prime basis" expansion ρ = Σ c_p(t) Φ_p e^(−λ_p t) is **[SUPERSEDED]**: functions on prime shells cannot represent a uniform start, and the time dependence is counted twice. The expansion of the dynamics is over the eigenvectors of the generator:

$$\rho(t) = \sum_{k=0}^{|\Lambda|-1} c_k \, \phi_k \, e^{-\lambda_k t}, \qquad 0 = \lambda_0 < \lambda_1 \le \lambda_2 \le \cdots$$

2.2 Absorbing Traps and the Dual-Helix Sampler

Proposition 2.3 (Absorbing Traps and Harmonic Measure). [PROVEN] Formalizes the selection rule of source §2 and the trap of source §4; also Fifteen Lemmas, Lemma 14 Part 2.

Make every prime-quadrance node absorbing and keep the rates of Definition 2.1 elsewhere, on a connected finite lattice with at least one prime node. From any start the mass converges to a distribution supported only on the prime nodes; its value at a prime node a is the probability of first entering the prime set at a, the harmonic measure of the prime set. Every transient mode decays at a rate of at least λ_min > 0.

Proof. Let L_TT be the generator restricted to the transient nodes and L_TA its rates into the traps. Every transient node has a path to a trap, so −L_TT is invertible and the transient mass decays at least as e^(−λ_min t). The mass absorbed from node t is row t of (−L_TT)⁻¹ L_TA, the hitting distribution of the prime set. ■

Remark (The 4 × 4 × 4 Lattice). [COMPUTED] With V = 0 and D = 1 the 25 prime nodes trap all the mass; the slowest transient mode decays at λ_min = 1.402, the mean absorption time from a uniform start is 0.295 in units of 1/D, and the final trap masses range from 0.031 to 0.045.

The Dual-Helix report samples candidates with ∂ρ/∂t = ∇·(D(Q)∇ρ) − V(Q)ρ, with D(Q) = 1/(1 + αχ(Q)) slowed by the TQF collinearity feature χ (BioChain AI 2026a, equation 23). The term −Vρ removes mass at rate V: it is a killing term, a soft trap.

Proposition 2.4 (Killing Concentrates but Does Not Select). [PROVEN] Formalizes BioChain AI (2026a), equation (23), on a lattice.

Let the killing rate be 0 on prime nodes and κ > 0 on the others. The total mass decays, and the normalized surviving mass converges to the left Perron eigenvector

of the generator minus diag(κ), which is strictly positive on every node of a connected lattice.

Proof. The generator minus a non-negative diagonal has non-negative off-diagonal entries and an irreducible pattern, so by Perron–Frobenius its leading eigenvalue is real and simple with a strictly positive eigenvector, and every other mode decays faster relative to it. ■

Remark (Killing Is Slower and Costlier). **[COMPUTED]** On the same lattice, killing rates κ = 1, 4, 16 and 64 on composite nodes give quasi-stationary prime-node masses 0.437, 0.575, 0.835 and 0.955, while the surviving mass decays at rates 0.563, 1.699, 2.649 and 2.911. Gibbs barriers reach higher concentration without destroying mass, and only absorbing traps give exact support.

2.3 The Infinite Lattice and Drift Without a Potential

Two results of the Fifteen Lemmas complete the picture: the sieve extends from a finite box to the whole Hurwitz lattice exactly when a series converges, and a drift along lattice lines with no potential cannot sieve at all.

Proposition 2.5 (The Sieve on the Infinite Hurwitz Lattice). **[PROVEN]** Fifteen Lemmas, Lemma 14 Part 3; the sharpness statement is new.

On the infinite Hurwitz lattice with a radial potential V(q) = V(N(q)), the Gibbs state of Proposition 2.2 exists exactly when the partition function converges:

$$Z_{\mathcal{H}}(\beta) = 1 + \sum_{n=1}^{\infty} 24\,\sigma_{\mathrm{odd}}(n)\,e^{-\beta V(n)} < \infty$$

For V(n) = α ln n this holds exactly when βα > 2, and every linear trap V(n) = κn with κ > 0 qualifies.

Proof. The shell of norm n holds 24σ_odd(n) Hurwitz integers (Proposition 4.1), which gives the series. For V = α ln n it is 24 Σ σ_odd(n) n^(−s) with s = βα, and as a Dirichlet series Σ σ_odd(n) n^(−s) = (1 − 2^(1−s)) ζ(s) ζ(s − 1). This converges for s > 2 and diverges at s = 2, where ζ(s − 1) has its pole and 1 − 2^(1−s) = 1/2; with positive terms it diverges for every s ≤ 2. A linear trap gives terms below 24 n² e^(−βκn). ■

Remark (Measured). **[COMPUTED]** The partial sums of 24 Σ σ_odd(n) n^(−s) to n = 1,000 and 4,000 are 150.2 and 177.5 at s = 2, still growing, and 35.57 and 35.59 at s = 3. The finite boxes of this paper are truncations; a sieve on the whole lattice needs a potential that confines at least as fast as (2/β) ln n.

Proposition 2.6 (Drift Without a Potential Does Not Sieve). **[PROVEN]** Fifteen Lemmas, Lemma 7 Part 3(e); the entropy rate uses Hoard (2026d), Proposition 4.8(4); formalizes the "collinear flow" of source §4.

On the torus (ℤ/4)³, give every forward step +e_k probability w_k/W and every reverse step −e_k probability ρ_k w_k/W, the same at every cell, with W = 2(w₁ + w₂ + w₃), 0 < ρ_k ≤ 1 and the remaining mass on the diagonal. The chain is doubly stochastic, its unique stationary state is uniform, and each forward edge carries the

stationary current $(1 - \rho_k)w_k/(64W)$, positive whenever $\rho_k < 1$. The chain produces entropy at the rate $\sigma = \Sigma_k (1 - \rho_k)\, w_k \ln(1/\rho_k)/W$ per step, positive whenever some $\rho_k < 1$.

Proof. Translation invariance gives every column the same multiset of entries as a row, so columns sum to 1 and the uniform vector is stationary; the steps generate the torus, so it is unique. The current on a forward edge is the forward flow minus the reverse flow, $(w_k - \rho_k w_k)/(64W)$. The entropy production of a stationary chain is $\sigma = \tfrac{1}{2} \Sigma (F_{ab} - F_{ba}) \ln(F_{ab}/F_{ba})$ with $F_{ab} = \mu_a K_{ab}$ (Schnakenberg 1976; Hoard 2026d, Proposition 4.8(4)); each of the 64 forward edges along axis k contributes $(1 - \rho_k)w_k/(64W)$ times $\ln(1/\rho_k)$. ■

Remark (Current Without Selection). **[COMPUTED]** With weights 1, 2 and 3 and valves 0.5, 0.7 and 0.9, the forward currents are 6.5×10^{-4}, 7.8×10^{-4} and 3.9×10^{-4} and the entropy production is 0.0494 per step, while the prime-node share stays 25/64 = 0.3906, the barrier-zero value of Table 5. A drift along lines moves mass around the lattice but cannot favour prime nodes. For drifts that vary from cell to cell the edge-wise statement fails, while the net flux through each cross-section stays positive in trials (companion, Problem 6).

Remark (Kolmogorov's Criterion). **[PROVEN]** The cubit installment proves for a three-state chain that the cycle current vanishes exactly when the products of the rates around the cycle agree in both directions, and that the entropy production is then zero (Hoard 2026d, Proposition 4.8(3) and (4)); Kolmogorov's criterion extends it to any finite chain: the stationary state carries no current on any edge exactly when the rate products agree in both directions around every cycle (Kolmogorov 1936). The two mechanisms of this section lie on opposite sides of it. Gibbs rates $k_{ab} = D\, e^{-(V_b - V_a)/2}$ have equal products around every cycle, since the potential differences telescope, so a sieve driven by a potential carries no current and produces no entropy in its stationary state. The drift of Proposition 2.6 has unequal products around each axis loop, $(w_k/W)^4$ against $(\rho_k w_k/W)^4$, so it carries a current and produces entropy (Appendix A, section Z). Selection comes only from the first kind and circulation only from the second: a potential concentrates mass on the primes at no stationary cost, while a pump pays continuously and concentrates nothing.

> **Editorial Note.** Barriers, killing and traps can concentrate mass on primes, and all three need the prime set as an input; drift alone cannot concentrate it at all. The Fifteen Lemmas name this the Circularity Boundary (Lemma 14 Part 4), and Section 3 shows why it is not a defect under the Dual-Helix principle of the series.

3 Geometry Proposes, Arithmetic Certifies

The sieve does not need to compute primes: under the Dual-Helix principle it proposes trap sites and an exact test certifies them, and the Engine's 30-node wheel gives the proposer a proven bias of 3.75, the largest of any wheel that fits the packet's 3-bit wheel field.

Status: [PROVEN] and [COMPUTED]. Sources: BioChain AI (2026a), §§1, 4 and 5, for the principle, the exactness theorem and the Oracle Bias Hypothesis; BioChain AI (2026b), §2, for the Helix A and Helix B roles; Hoard (2026a), §5, read as $\mathbb{Z}/30$ (Hoard 2026d, §4.6), most accurate method; Hoard (2026d), Theorem 4.9, for the base-6 wheel; Fifteen Lemmas, Lemma 14 Part 4 (the Circularity Boundary) and Lemma 4 Part 5 (the Legendre split).

3.1 The Sieve as a Helix A Oracle

In the Dual-Helix report, Helix A is a continuous geometric oracle that ranks candidates, and Helix B reconstructs exact residues and applies a GCD test; every reported factor is exact whatever the oracle does (BioChain AI 2026a, Theorem 4.1). The prime sieve has the same shape. The Fokker-Planck dynamics of Section 2 decide where mass goes; an exact primality test decides which nodes are traps. The Fifteen Lemmas prove the boundary between the two: in the Gibbs limit and the trap limit alike, the limiting support is the set the potential or the trap rule singles out, so a sieve that uses a primality test displays and samples a prime set rather than computing it, and its dynamics belong in Helix A (Lemma 14 Part 4).

> **Proposition 3.1 (Certified Traps Are Exact).** **[PROVEN]** The sieve analogue of BioChain AI (2026a), Theorem 4.1.
>
> *If a trap is placed at a node only when an exact test certifies its norm prime, then every node that absorbs mass has prime norm, whatever the potential, drift, diffusion tensor or numerical precision of the dynamics.*
>
> *Proof. Mass is absorbed only at trap nodes, and every trap node is certified. The dynamics decide which certified node absorbs the mass, never whether a node is certified.* ■

> **Proposition 3.2 (The Wheel Is a Proven Oracle Bias).** **[PROVEN]** Uses Hoard (2026a), §5, read as $\mathbb{Z}/30$ (Hoard 2026d, §4.6).
>
> *Let a candidate $N > 5$ pass the 30-node wheel when $N \bmod 30$ is one of the 8 units of $\mathbb{Z}/30$. Every prime above 5 passes, and the density of primes among passing integers tends to $30/8 = 3.75$ times their density among all integers.*
>
> *Proof. A prime above 5 is coprime to 30. By Dirichlet's theorem the primes are equidistributed among the $\varphi(30) = 8$ unit classes, while passing integers make up 8/30 of all integers, so the ratio of densities tends to 30/8.* ■

Remark (The Bias Measured). **[COMPUTED]** Among the integers from 6 to 10^7 the prime density is 0.06646; among those passing the wheel it is 0.24922, a ratio of 3.7500. In the notation of BioChain AI (2026a), Assumption 5.1, with B the event that a candidate is prime, U the uniform candidate distribution and D the wheel-filtered one, $\Pr_D(B) = \beta \Pr_U(B)$ with $\beta \to 3.75$. The Oracle Bias Hypothesis therefore holds for the prime task, as a theorem. For factoring it remains **[CONJECTURAL]**, and nothing here bears on it.

Remark (The Legendre Split). **[PROVEN]** The 8 wheel units split by the Legendre symbol mod 5 into $B_+ = \{1, 11, 19, 29\}$ and $B_- = \{7, 13, 17, 23\}$ (Fifteen Lemmas, Lemma 4 Part 5). By Dirichlet's theorem each half carries half of the primes above 5. The class of a candidate is one of the 8 values of the Einstein Tile's 3-bit wheel field (Section 8.1), and the Legendre symbol reads its half without a lookup.

> **Proposition 3.3 (The Wheel That Fits the Tile).** **[PROVEN]** Extends Hoard (2026d), Theorem 4.9 and its Remark on the 30-node wheel.
>
> *For the primorial wheels modulo 6, 30 and 210, the unit classes number 2, 8 and 48, the passing integers have density 1/3, 4/15 and 8/35, and the oracle bias tends to 3, 3.75 and 4.375. The 30-node wheel tests 20 % fewer candidates than the base-6*

wheel, and it is the largest primorial wheel whose unit classes fit a 3-bit field, which they fill exactly.

Proof. The unit classes number φ(6) = 2, φ(30) = 8 and φ(210) = 48. As in Proposition 3.2, the primes above the largest prime factor of the modulus m are equidistributed among the φ(m) unit classes, so the bias tends to m/φ(m). The ratio of candidate densities is (8/30)/(1/3) = 0.8. Since 8 = 2³ and 48 > 8, ℤ/30 is the largest primorial modulus whose units index into 3 bits. ■

Table 6: Primorial wheels as oracles for the prime task.

Modulus	Unit classes	Candidates tested	Bias limit	Measured, integers to 10^7	Field width
6, the base-6 wheel (Hoard 2026d)	2	1/3 = 0.333	3	3.000	1 bit
30, the Engine's 30-node wheel	8	4/15 = 0.267	3.75	3.750	3 bits
210	48	8/35 = 0.229	4.375	4.375	6 bits

Remark (Why ℤ/30). [COMPUTED] A mod-210 wheel would raise the bias by a further 17 % but needs a 6-bit field. On the Einstein Tile, whose wheel field has 3 bits (Section 8.1), ℤ/30 is the best wheel available, and its eight classes use every code of the field. This is the precise sense in which the Engine's wheel is the most accurate method for the packet while not being the best wheel in general, as the cubit installment already noted (Hoard 2026d, §4.6; Appendix A, section Z).

3.2 Exact Certification at Every Scale

Helix B needs an exact primality test at the scale where traps are placed. Table 7 lists the test and its arithmetic width at the three scales this paper uses.

Table 7: Exact primality certification by scale.

Scale	Exact test	Arithmetic width	Cost
Form 1 packet core, N ≤ 16,129	Lookup in a bitmap of the 1,877 primes up to 16,129	14-bit index	One memory read; the table is 2,017 bytes
64-bit word, N < 2^{64}	Miller–Rabin with the twelve bases 2, 3, 5, …, 37, deterministic below 3.3×10^{24} (Sorenson and Webster 2017)	128-bit products	At most 12 modular exponentiations
Horizon, Q ≈ 10^{12}	The same test	80-bit products	As above; the TQF needs two 128-bit products in Gram form (Table 13)

Combined with the wheel, 73.3 % of integer candidates and 65.7 % of Form 1 cores skip the test altogether (Section 8.2).

> **Editorial Note.** Correctness comes from Helix B, so Helix A may use any geometric heuristic: a potential, a drift, the golden torus of Section 5 or the hyperbolic tree of Section 7. Whether a potential defined without any primality test could still place its minima exactly on the primes is recorded, with a falsifiable test, as Problem 9 of the companion; the sieve does not depend on the answer.

4 The Four-Dimensional Setting

The sieve belongs in four dimensions: the Hurwitz order carries every prime, a quarter of the primes never occur in three dimensions, and every integer quaternion has a canonical, frame-free form.

Status: [ESTABLISHED] and [PROVEN]. Sources: source note §3; BioChain AI (2026h) for the parallelizable 3-sphere and the scalar key; Conway and Smith (2003); Fifteen Lemmas, Lemmas 5, 8 and 13 and §6.1; Hoard (2026d), Proposition 4.6(3), for stated sign conventions.

4.1 The Hurwitz Order and the Three-Square Obstruction

A Hurwitz quaternion $H = a + bi + cj + dk$ has four integer or four half-integer components; its norm $N(H) = a^2 + b^2 + c^2 + d^2 = H\bar{H}$ is a non-negative integer. The Hurwitz quaternions form a ring with left and right division with remainder, and its 24 units form the binary tetrahedral group 2T.

> **Proposition 4.1 (No Inert Primes; Norm Counts). [ESTABLISHED]** Formalizes source §3.
>
> *(1) Every rational prime p is the norm of a Hurwitz integer π, so $p = \pi\bar{\pi}$ and no prime is inert. (2) The number of Hurwitz integers of norm n is 24 times the sum of the odd divisors of n; for an odd prime p it is 24(p + 1). (3) A Hurwitz integer is irreducible exactly when its norm is prime.*

Part (1) is Lagrange's four-square theorem; parts (2) and (3) are Hurwitz's (Conway and Smith 2003). An integer quaternion of prime norm is primitive, since $g = \gcd(a, b, c, d)$ gives g^2 dividing p **[PROVEN]**; the note's stability condition is therefore automatic on prime shells.

> **Theorem 4.2 (Three-Square Obstruction). [ESTABLISHED]** Gives the note's "Prime Conflict in 3D" a precise form.
>
> *A positive integer is a sum of three squares exactly when it is not of the form $4^a(8b + 7)$ (Legendre). Hence no prime $p \equiv 7 \pmod 8$ is the quadrance of a point of $\mathbb{Z}^3$.*

Remark (Missing Primes). [COMPUTED] Of the 17,984 primes below 200,000, 4,511 (25.08 %) are congruent to 7 mod 8 and none is a sum of three squares. The 4 × 4 × 4 lattice with coordinates 0 to 3 has 25 prime-quadrance nodes and misses 7 and 23; the 4 × 4 × 4 × 4 lattice has 94 prime nodes and carries every prime up to 23. Centred on the origin, the Hurwitz points of the box $[-2, 2]^4$ reach exactly the primes 2, 3, 5, 7 and 13; 11 needs a coordinate of absolute value 3 (Fifteen Lemmas, Lemma 5 Part 4) **[PROVEN]**.

Remark (Why Four Dimensions, from the Companion Papers). [ESTABLISHED] The voxel-alignment review rests on two facts it states informally (BioChain AI 2026h). The only parallelizable spheres are S^0, S^1, S^3 and S^7 (Bott and Milnor 1958; Kervaire 1958), and the only normed division algebras over the reals have dimension 1, 2, 4 and 8 (Hurwitz 1898). These are exactly the four columns of that paper's scaling table, and S^3 is the unit sphere of the quaternions. Its "locked state" w = 0 is the 2-sphere of unit pure quaternions, where the hairy-ball theorem forbids a nowhere-vanishing tangent field; Section 7.6 identifies it with the boundary at infinity of hyperbolic space.

Remark (Quaternion Matrices). **[ESTABLISHED]** The 4 × 4 matrix L(q) of left multiplication by q = w + xi + yj + zk satisfies $L(q)^T L(q) = N(q)I$ and $L(q) + L(q)^T = 2wI$, so it is a scaled orthogonal matrix with $\det L(q) = N(q)^2$ (Fifteen Lemmas, Lemma 8, Step 2). Section 9.3 uses it.

4.2 Canonical Forms and Unique Factorization

Two agents that store the same lattice point in frames differing by a unit see different coordinates. A canonical associate removes that difference without any shared reference.

> **Proposition 4.3 (Canonical Associate).** **[PROVEN]** Address normal form.
>
> *Let q be an integer quaternion with N(q) ≡ 1 (mod 4). Exactly one of its eight associates uq, with u ∈ {±1, ±i, ±j, ±k}, has odd positive real part.*
>
> *Proof. Squares are 0 or 1 mod 4, so N(q) ≡ 1 forces exactly one odd coordinate. The real part of uq runs over ±a, ±b, ±c and ±d as u runs over the eight units, each value once. The odd coordinate is non-zero, so it appears with positive sign exactly once.* ■

For other norms the canonical associate is defined as the lexicographic maximum of the 24 Hurwitz associates **[DEFINITION]**. The unit that carries q to its canonical form is the frame of q relative to the standard.

Scope of the canonical associate. **[DEFINITION]** It normalizes lattice points used as addresses. A quaternion that encodes a rotation must not be replaced by uq, since uq rotates space differently. For rotations the Einstein Tile's Form 1 instead uses sign normalization, the first non-zero coordinate positive, which leaves $x \mapsto Qx\bar{Q}/N(Q)$ unchanged and moves the spinor sign to the cycle-step field (Fifteen Lemmas, §6.1). The two normal forms answer different questions: which lattice point, and which rotation. Either sign convention for rotations is valid once it is stated; the cubit installment makes the same point for the logarithm of a unit quaternion, where SLERP implementations replace q by −q when Re q < 0 (Hoard 2026d, Proposition 4.6(3)).

> **Theorem 4.4 (Unique Factorization Modelled on a Norm Order).** **[ESTABLISHED]** Conway and Smith (2003), Chapter 5.
>
> *Let H be a primitive Hurwitz integer of norm $n = p_1 p_2 \ldots p_k$, with the primes taken in a fixed order. Then $H = \pi_1 \pi_2 \ldots \pi_k$ with $N(\pi_i) = p_i$, and the factorization is unique up to unit migration, the replacement of $\pi_i \pi_{i+1}$ by $(\pi_i u)(u^{-1}\pi_{i+1})$.*

The ordered sequence of prime-norm factors, taken up to unit migration, is therefore a canonical address of a primitive lattice point: it depends only on the point and the agreed order of primes, not on any frame. Section 7 makes it explicit for norms that are powers of 5.

> **Editorial Note.** The sieve, its traps and its canonical forms all live in the Hurwitz order of four-dimensional space. Section 5 adds a sharper reason drawn from the golden torus of the series.

5 Prime Nodes on the Golden Clifford Torus

The golden Clifford torus is the series' search surface, and it contains no lattice point, so no prime trap can sit on it; its Fibonacci approximants do carry lattice points, whose norms are Lucas numbers, and the prime ones beyond 3 are exactly primes that three dimensions cannot hold.

Status: [PROVEN] and [COMPUTED]. Sources: BioChain AI (2026c), §§2–4, for the torus and its delta encoding; BioChain AI (2026a), §3.7, which uses the torus as a decorrelating search surface; Fifteen Lemmas, Lemmas 8 and 11.

5.1 The Torus and the Delta Encoding

The golden Clifford torus is the set of unit quaternions with radius ratio $r_1/r_2 = \varphi$ (BioChain AI 2026c, §2.1), so that $r_1^2 = (5 + \sqrt{5})/10$ and $r_2^2 = (5 - \sqrt{5})/10$. Writing its points in complex form makes the delta encoding transparent:

$$q(\theta_1, \theta_2) = r_1 e^{i\theta_1} + r_2 e^{i\theta_2} j, \qquad r_1^2 + r_2^2 = 1, \quad r_1/r_2 = \varphi$$

Proposition 5.1 (Torus-Preserving Delta). [PROVEN] The delta encoding of BioChain AI (2026c), §3.1, in torus-preserving form.

For real α and β, $e^{i\alpha}\, q(\theta_1, \theta_2)\, e^{i\beta} = q(\theta_1 + \alpha + \beta, \theta_2 + \alpha - \beta)$. A phase step $(\Delta\theta_1, \Delta\theta_2)$ is therefore applied exactly, on the torus, by $\alpha = (\Delta\theta_1 + \Delta\theta_2)/2$ and $\beta = (\Delta\theta_1 - \Delta\theta_2)/2$.

Proof. $e^{i\alpha}$ commutes with $e^{i\theta}$, and $j\, e^{i\beta} = e^{-i\beta}\, j$. Hence $e^{i\alpha}\, q\, e^{i\beta} = r_1\, e^{i(\theta_1 + \alpha + \beta)} + r_2\, e^{i(\theta_2 + \alpha - \beta)}\, j$. ■

Appendix A, section L, confirms the two-sided step exact in 1,000 random trials.

Remark (Covering the Torus). [PROVEN] A sampler that runs a stream $\theta_1 = \omega_1 t$, $\theta_2 = \omega_2 t$ covers the torus densely exactly when ω_1/ω_2 is irrational (Kronecker); the Fifteen Lemmas prove it with Weyl's criterion and show that the golden radii alone do not force it (Lemma 11 Parts 2 and 5). The choice $\omega_1/\omega_2 = \varphi$ makes the stream dense, while a rational winding closes: the (2, 3) trefoil $K(s) = (r_1 e^{2is}, r_2 e^{3is})$ lies on the golden Clifford torus and closes at $s = 2\pi$ (Fifteen Lemmas, Lemma 8, Proposition 6.5). The companion records how this replaces the source's one-sided delta and its ergodicity claim (Hoard 2026h, Section 4.1).

5.2 No Lattice Point on the Golden Torus

Theorem 5.2 (The Golden Torus Avoids the Lattice). [PROVEN] Bears on the golden solitons of BioChain AI (2026c), §4.1.

No point of $\mathbb{Q}^4$ other than 0 lies on a Clifford torus with radius ratio φ, at any scale.

Proof. A point (a, b, c, d) on such a torus satisfies $a^2 + b^2 = \varphi^2(c^2 + d^2)$. If $c^2 + d^2 = 0$ then $a^2 + b^2 = 0$ and the point is 0. Otherwise $\varphi^2 = (a^2 + b^2)/(c^2 + d^2)$ would be rational, but $\varphi^2 = (3 + \sqrt{5})/2$ is irrational. ■

A prime trap therefore cannot be a node on the torus itself. The torus is a search surface, which is how the Dual-Helix report uses it (BioChain AI 2026a, §3.7); exact nodes must come from a nearby rational torus.

5.3 Fibonacci Approximants and Lucas Primes

The Fibonacci ratios F_{2m+1}/F_{2m−1} converge to φ², and each defines a rational Clifford torus that does carry lattice points.

Theorem 5.3 (Lattice Points on the Fibonacci Tori). [PROVEN]

Let q_m = F_m + F_{m+1} i + F_{m−1} j + F_m k. Then (1) q_m lies on the Clifford torus with radius ratio squared F_{2m+1}/F_{2m−1}, and N(q_m) = L_{2m}, the Lucas number; (2) if L_{2m} is prime then m is a power of 2; (3) for m = 2^k with k ≥ 1, L_{2m} ≡ 7 (mod 8), so these primes never occur as quadrances in three dimensions.

Proof. (1) The identities F_m² + F_{m+1}² = F_{2m+1}, F_{m−1}² + F_m² = F_{2m−1} and F_{2m+1} + F_{2m−1} = L_{2m} give both claims. (2) If m = 2^a b with b odd and b > 1, then 2m/2^{a+1} = b is odd, so L_{2^{a+1}} divides L_{2m}, and 3 ≤ L_{2^{a+1}} < L_{2m}. (3) For even m, L_{2m} = L_m² − 2. L_m is odd because 3 does not divide m, so L_m² ≡ 1 and L_{2m} ≡ 7 (mod 8); Theorem 4.2 applies. ■

Remark (The First Approximants). [COMPUTED] Table 8 lists q_m for m = 1 to 8. The norms are prime for m = 1, 2, 4 and 8, giving 3, 7, 47 and 2207, and the distance of the radius ratio from φ² shrinks by a factor of about φ⁴ ≈ 6.85 per step. Among even n ≤ 800, L_n is prime only for n = 2, 4, 8 and 16.

Table 8: Lattice points on the Fibonacci approximants of the golden Clifford torus.

m	q_m	Norm L_{2m}	Prime	Norm mod 8	Distance of the ratio from φ²
1	(1, 1, 0, 1)	3	yes	3	6.18×10^{-1}
2	(1, 2, 1, 1)	7	yes	7	1.18×10^{-1}
3	(2, 3, 1, 2)	18	no	2	1.80×10^{-2}
4	(3, 5, 2, 3)	47	yes	7	2.65×10^{-3}
5	(5, 8, 3, 5)	123	no	3	3.87×10^{-4}
6	(8, 13, 5, 8)	322	no	2	5.65×10^{-5}
7	(13, 21, 8, 13)	843	no	3	8.24×10^{-6}
8	(21, 34, 13, 21)	2207	yes	7	1.20×10^{-6}

Whether infinitely many L_{2^k} are prime is open (companion, Problem 1); L_{32} = 4,870,847 = 1087 × 4481 is the first composite one.

Remark (The Discrete Golden Circle). [ESTABLISHED] The one-dimensional counterpart is the Three-Distance Theorem: the points k/φ mod 1 for 0 ≤ k < N leave at most three gap lengths, and when N = F_m exactly two, in ratio φ: F_{m−1} gaps of length φ^{−(m−2)} and F_{m−2} of length φ^{−(m−1)} (Sós 1958; Fifteen Lemmas, Lemma 11 Part 3). In both settings Fibonacci truncation is where the golden structure becomes exactly finite: on the circle as a two-gap partition, on the torus as the lattice points q_m.

Editorial Note. The golden torus and the prime sieve meet as the Dual-Helix principle predicts: the irrational torus is the Helix A search surface, and the Fibonacci lattice nodes are where Helix B can certify. The approximants also give the sharpest reason yet for four dimensions: the prime nodes 7, 47 and 2207 are invisible to any three-dimensional lattice.

6 Exact Arithmetic and Closure

Exact Hurwitz arithmetic removes drift from the sieve's lattice evolution outright, and in exact arithmetic a periodic schedule of deltas returns to the identity within 12 repetitions or never.

Status: [PROVEN]. Sources: Fifteen Lemmas, Lemma 13, whose proofs are imported; Hoard (2026a), §I.1, for $W_\Sigma = 60$, source only.

Theorem 6.1 (Exact Deltas Close Within 12 Repetitions or Never). [PROVEN] Fifteen Lemmas, Lemma 13 Parts 1 and 2; the corollary for schedules is new.

Let w be a product of rational unit quaternions. If w has finite order, that order is 1, 2, 3, 4 or 6, so $w^{12} = 1$. Otherwise no power of w is 1; for example the powers of $(3 + 4i)/5$ have exact denominators 5^n. In particular the elements of order 5, 8 and 10 in the binary octahedral and icosahedral groups 2O and 2I are irrational and cannot occur in exact arithmetic, and a periodic schedule of rational deltas that returns at all returns within 12 repetitions.

Proof. A finite-order unit quaternion is $\cos\theta + \hat{u}\sin\theta$ with θ a rational multiple of π, and its real part $\cos\theta$ is rational. By Niven's theorem $\cos\theta \in \{0, \pm 1/2, \pm 1\}$, so the order divides 12. For $(3 + 4i)/5$: $3 + 4i = (2 + i)^2$ in the Gaussian integers, and the prime $2 - i$ never divides a power of $2 + i$, so 5 never divides both components of $(3 + 4i)^n$. ■

Proposition 6.2 (Exact Arithmetic Has No Drift). [PROVEN] Uses Fifteen Lemmas, Lemma 13 Part 3.

Integer and Hurwitz quaternions form rings closed under multiplication, and $N(pq) = N(p)N(q)$ exactly. A lattice evolution computed in integer or half-integer arithmetic accumulates no rounding error, and the sieve needs no periodic reset. A product of k integer quaternions with coordinates below 2^b has coordinates below $2^{k(b+2)-2}$, so exactness lasts as long as the word width allows; Section 7.2 gives a much sharper bound for the norm-5 generators.

Proof. Closure and the multiplicativity of the norm are ring identities. Each coordinate of a product AB is a signed sum of four products of coordinates, so its largest coordinate is at most 4 times the product of the largest coordinates of A and B; induction over k factors gives the bound. ■

Rational, rather than integer, evolution does drift in width: by Theorem 6.1 its denominators grow without bound unless the schedule has finite order. Floating-point evolutions still need a reset; the right one is a renormalization or a snap to the nearest lattice point, at any interval. The companion examines the 720-step cycle that the symmetry-folding paper proposes for this purpose (Hoard 2026h, Section 4.4).

Remark (The Engine's 60). [CORRESPONDENCE] The exponent of 2I is 60, the Engine's total winding W_Σ (Hoard 2026a, §I.1); 2I is also SL(2, 5), of order 120, and its rotation image A_5 has order 60 (Section 7.5). By Theorem 6.1, 2I cannot be realized exactly, so the equality is a numerical correspondence; no mechanism is claimed.

7 Procedural Generation in Hyperbolic Space and the Standardization of Cubic Integers

The six quaternions 1 ± 2i, 1 ± 2j, 1 ± 2k are the six steps of the cubic lattice made non-commuting. Their words grow a 6-regular hyperbolic tree whose abelian shadow is the cubic lattice; every address factors back to a unique word; and reduction modulo a prime turns the tree into a finite Ramanujan graph on which agents can meet.

Status: [ESTABLISHED], [PROVEN] and [COMPUTED]. Sources: BioChain AI (2026d), §3 and §8, horizon 2, for hyperbolic branching and the shift question; BioChain AI (2026a), §3.3, and BioChain AI (2026h) for the Lorentz lift and the scalar key; Hoard (2026a), §III.1, for $C_3 \times C_2$, source only; Lubotzky, Phillips and Sarnak (1988); Serre (1980); Fifteen Lemmas, Lemmas 8, 9 and 13 and Theorems 7.4, 7.8 and 7.19.

7.1 The Cubic Lattice as an Abelian Shadow

Definition 7.1 (The Norm-5 Generators). [DEFINITION]

$S_5 = \{1 + 2i, 1 - 2i, 1 + 2j, 1 - 2j, 1 + 2k, 1 - 2k\}$. Each element has norm 5, and its conjugate is its inverse up to the scalar 5. The shadow map σ sends 1 ± 2i, 1 ± 2j, 1 ± 2k to the cubic steps $\pm e_1$, $\pm e_2$, $\pm e_3$ of $\mathbb{Z}^3$, and a word to the sum of its steps.

Theorem 7.2 (Free Generation and Unique Addresses). [ESTABLISHED] Lubotzky, Phillips and Sarnak (1988); **[COMPUTED]** for L ≤ 5.

(1) Modulo scalars, S_5 generates a free group of rank 3, whose Cayley graph is the 6-regular tree. (2) The reduced words of length L, those never followed by their own conjugate, give 6 × 5^{L−1} distinct quaternions, and up to sign these are exactly the primitive quaternions of norm 5^L with odd real part and even imaginary parts. (3) Hence each such quaternion has a unique word, recovered by repeated right division: exactly one g in S_5 leaves $q\bar{g}$ divisible by 5.

Appendix A, section P, checks (2) against a direct enumeration for every L ≤ 5 and checks (3) on 2,000 random words of length up to 12, all recovered exactly.

Two results of the Fifteen Lemmas place Theorem 7.2. First, freeness begins with each letter: 1 + 2i is a Gaussian prime not associate to its conjugate, so by the argument of Fifteen Lemmas, Lemma 13 Part 1, no power of it is real, and no one-letter word ever closes **[PROVEN]**. Second, generic tuples of unit quaternions generate free groups (Epstein 1971; Fifteen Lemmas, Theorem 7.19); S_5 is an explicit integral instance in which every relation-free word is also exactly computable. The contrast with units is sharp. A product of Hurwitz units forgets its factors: the 13,824 three-step chains of units give only 24 products, 576 chains each, so no scheme rebuilds a unit chain's history from its product (Fifteen Lemmas, §7.2). A product of norm-5 generators remembers its factors, because each step multiplies the norm by a prime.

In $\mathbb{Z}^3$ the six steps commute, so a walk of length L reaches only $O(L^3)$ points. In the tree they do not commute, and the number of addresses grows as 5^L: the tree is a discrete hyperbolic space, and $\mathbb{Z}^3$ is its image under abelianization **[PROVEN]**, since the abelianization of the free group on three generators is $\mathbb{Z}^3$ and σ is that map.

Table 9: Tree addresses against their cubic-lattice shadows.

Depth L	Tree addresses	Cubic-lattice shadows	Addresses per shadow	Distinct closed words at the origin
1	6	6	1.0	0
2	30	18	1.7	0
3	150	44	3.4	0
4	750	85	8.8	24
5	3,750	146	25.7	0
6	18,750	231	81.2	360
7	93,750	344	272.5	0
8	468,750	489	958.6	6,120

7.2 Generation by Shift and Add

Proposition 7.3 (Generator Steps Need No Multiplier). **[PROVEN]** Answers BioChain AI (2026d), §8, horizon 2, for S_5.

For $u \in \{\pm i, \pm j, \pm k\}$, $(1 + 2u)q = q + 2(uq)$ and $q(1 + 2u) = q + 2(qu)$. Multiplying by a unit u permutes the four coordinates of q and changes signs. Each generation step is therefore one signed permutation, one left shift by one bit and one four-lane addition.

Proof. Distributivity gives the first identities. Left multiplication by i sends $a + bi + cj + dk$ to $-b + ai - dj + ck$, and j and k act likewise. ■

The norm grows by 5 per step and each coordinate is at most 5^{L/2} in absolute value. A signed byte therefore holds every address to depth 6, 18,750 of them at depth 6, and a 64-bit signed coordinate holds every address to depth 54 **[COMPUTED]**.

7.3 Why Coordinates Cannot Align Hyperbolic Agents

Table 9 states the interoperability problem exactly. Two agents that generate content in the tree but compare only cubic coordinates cannot tell apart, on average, 958 different addresses at depth 8, and 6,120 distinct closed words return to the same origin. Agents that compare floating-point hyperbolic coordinates fail the other way: in the hyperboloid model a point at distance d has coordinates of size about e^d/2, so a fixed number of significant bits resolves positions ever more coarsely with depth **[ESTABLISHED]**. Exact integer quaternions avoid both failures: the address is the quaternion, and by Theorem 7.2 the quaternion determines its word.

7.4 A Finite Meeting Space by Congruence

A tree has no finite place to meet. Reducing the generators modulo a second prime folds the tree into a finite graph that keeps its expansion.

Theorem 7.4 (Ramanujan Meeting Graphs). **[ESTABLISHED]** Lubotzky, Phillips and Sarnak (1988).

For a prime $q \equiv 1 \pmod 4$, $q \neq 5$, map S_5 into PGL(2, $\mathbb{F}_q$) using a square root of -1 mod q. The Cayley graph $X(5, q)$ is 6-regular, on PSL(2, $\mathbb{F}_q$) when 5 is a square mod q and on PGL(2, $\mathbb{F}_q$), bipartite, otherwise. Every eigenvalue other than ± 6 has absolute value at most $2\sqrt{5} \approx 4.472$.

Remark (Against the Cubic Torus). [COMPUTED] Table 10 compares X(5, 13) and X(5, 29) with cubic tori of nearly the same size. Both families are 6-regular, so the comparison isolates the effect of non-commuting steps. The mixing time is the first time at which the sieve's diffusion, the continuous-time walk with generator $A - 6I$, is within 1 % of uniform in total variation from a point start.

Table 10: The sieve's diffusion on Ramanujan graphs and on cubic tori of the same degree.

Graph	Nodes	Second eigenvalue	Spectral gap	Eccentricity	Mixing time (1 %)
X(5, 13), on PGL(2, 13)	2,184	4.2497	1.7503	7	3.25
Cubic torus 13^3	2,197	5.7709	0.2291	18	20.15
X(5, 29), on PSL(2, 29)	12,180	4.4420	1.5580	8	3.80
Cubic torus 23^3	12,167	5.9258	0.0742	33	62.15

Proposition 7.5 (Mixing in Logarithmic Time). [ESTABLISHED] Levin, Peres and Wilmer (2017), Chapters 12 and 20.

For the continuous-time walk on a regular graph with n nodes and spectral gap γ, the distance to uniform from any start is at most $\frac{1}{2}\sqrt{n}\, e^{-\gamma t}$. On $X(5, q)$, $\gamma \geq 6 - 2\sqrt{5} \approx 1.528$ for every q, so the mixing time grows like $\log n$; on the cubic torus m^3, $\gamma = 2(1 - \cos(2\pi/m)) \approx 4\pi^2/m^2$, and it grows like $n^{2/3}$.

Growing the graph 5.6-fold raises the mixing time from 3.25 to 3.80 on the Ramanujan graphs and from 20.15 to 62.15 on the tori.

Remark (A Second Hyperbolic Cube Structure). [ESTABLISHED] and **[PROVEN]**. The Fifteen Lemmas give a geometric counterpart to the norm-5 tree. A regular honeycomb {p, q, r} is spherical, Euclidean or hyperbolic as $\sin(\pi/p)\sin(\pi/r)$ exceeds, equals or falls below $\cos(\pi/q)$; for cubes, {4,3,3} gives 0.612, the cubic tiling {4,3,4} gives 0.5, and {4,3,5} gives 0.416, the compact hyperbolic honeycomb in which five cubes meet at each edge with dihedral angles of 72° (Coxeter 1973; Fifteen Lemmas, Theorem 7.4). In the Klein model its cell is the cube $[-a, a]^3$ with $a^2 = \sqrt{5} - 2 = \varphi^{-3}$, and the boosts to its six face-neighbours have $\cosh \vartheta = \varphi$ (Fifteen Lemmas, Lemma 8, Proposition 6.4). Table 11 compares the two structures.

Table 11: Two hyperbolic structures over the cube.

Property	Norm-5 tree (Section 7.1)	{4,3,5} honeycomb
Cells	Nodes of the 6-regular tree	Cubes, five around each edge
Steps	$1 \pm 2i$, $1 \pm 2j$, $1 \pm 2k$	Boosts across the six faces, $\cosh \vartheta = \varphi$
Arithmetic	Integers; shift and add	The number field $\mathbb{Q}(\sqrt{\varphi})$, of degree 4
Growth	$6 \times 5^{L-1}$ addresses at depth L	Exponential in hyperbolic radius
Finite quotient	Ramanujan graphs X(5, q)	Not used here
Role in the standard	Addresses and the meeting space	Geometry and rendering of cells

The tree is exact in integers and therefore carries addresses; the honeycomb carries the geometry of cubes, exactly but over $\mathbb{Q}(\sqrt{\varphi})$. The two are complementary readings of "procedurally generated cubes in hyperbolic space" **[DEFINITION]**.

7.5 The Modular Group, the Engine's Channels and the Hurwitz Units

Proposition 7.6 (Folding the Modular Group by Primes). **[ESTABLISHED]** Serre (1980); **[COMPUTED]** order profiles.

(1) PSL(2, $\mathbb{Z}$) is the free product $C_2 * C_3$, and its abelianization is $C_2 \times C_3 \cong \mathbb{Z}/6$. (2) SL(2, $\mathbb{F}_3$) is the binary tetrahedral group 2T, the 24 Hurwitz units, and SL(2, $\mathbb{F}_5$) is the binary icosahedral group 2I, of order 120, whose image PSL(2, $\mathbb{F}_5$) $\cong A_5$ has order 60. (3) In SL(2, $\mathbb{Z}$) the generators S and ST have orders 4 and 6, the orders of the Hurwitz unit i and of the Engine's tier rotation h.

The Engine's six channels form $C_3 \times C_2$ (Hoard 2026a, §III.1). By (1), that finite group is the abelian shadow of the hyperbolic modular group, as $\mathbb{Z}^3$ is the shadow of the norm-5 tree; the abelianization of a free product is the product of the abelianizations **[PROVEN]**. By (2), reducing the modular group modulo 3 lands on the sieve's unit group, and modulo 5 on a group whose exponent is the Engine's 60. Appendix A, section K, confirms that S and T generate both finite groups and that their element-order profiles match 2T and 2I exactly. That the Engine's constants reappear here is recorded, not explained **[CORRESPONDENCE]**.

7.6 The Lorentz Lift and the Scalar-Zero Boundary

Proposition 7.7 (The Scalar Key Is the Ideal Boundary). **[PROVEN]** Connects BioChain AI (2026a), Definition 3.6, with BioChain AI (2026h).

For a unit quaternion $q = w + v$ with $w > 0$, the Lorentz lift $\tilde{q} = (1/w, v/w)$ lies on the hyperboloid $t^2 - x^2 - y^2 - z^2 = 1$. As $w \to 0$ from above, $\tilde{q}$ runs off to the boundary at infinity, which is the 2-sphere of directions $v/|v|$.

Proof. $t^2 - |v|^2/w^2 = (1 - |v|^2)/w^2 = w^2/w^2 = 1$. *As $w \to 0$, $|v| \to 1$ and every coordinate of $\tilde{q}$ diverges along the direction of v.* ■

The voxel-alignment review's "locked state" w = 0, where the hairy-ball theorem bites, is therefore the ideal boundary of hyperbolic 3-space, and its "scalar key" w > 0 is what places a state inside it (BioChain AI 2026h). The rational hyperbolic quadrance $Q_h = \langle \tilde{p}, \tilde{q} \rangle^2 - 1$ of the Dual-Helix report is invariant under every Lorentz transformation, so it compares lifted states from any frame **[ESTABLISHED]**. The lift itself comes from Hoard (2026c).

Proposition 7.8 (The Ghost Map). **[PROVEN]** Fifteen Lemmas, Lemma 8, Proposition 6.2, and Theorem 7.8.

Compose the rational Cayley map $C_{rat}(p) = (1 - \|p\|^2, 2p)/(1 + \|p\|^2)$ with the Lorentz lift of Proposition 7.7. The result

$$G(\mathbf{p}) = \left(\frac{1 + \|\mathbf{p}\|^2}{1 - \|\mathbf{p}\|^2}, \frac{2\mathbf{p}}{1 - \|\mathbf{p}\|^2} \right)$$

is a bijection from the open unit ball onto the hyperboloid sheet, an isometry from the Poincaré ball model, rational on rational points, and equivariant under every

rotation. Between two projected states, $Q_h = \langle G_i, G_j\rangle^2 - 1 = \sinh^2 d_H(G_i, G_j)$, a rational, symmetric kernel that vanishes only when the states coincide.

So Q_h, used heuristically in the Dual-Helix report, has an exact meaning: it is a function of hyperbolic distance alone. Appendix A, section X, checks the distance formula exactly on 500 pairs of rational points. The projection of quaternion states into hyperbolic 3-space proposed by the symmetry-folding paper (BioChain AI 2026d, §3) is realized exactly by the lift and the ghost map; its group-theoretic form is the inclusion $SU(2) \subset SL(2, \mathbb{C})$ as the stabilizer of a base point of hyperbolic space (Fifteen Lemmas, Lemma 8, Proposition 6.3).

7.7 The Standard for Procedurally Generated Cubic Integers

Definition 7.9 (Standard Procedural Integer). [DEFINITION]

A procedurally generated lattice point is exchanged as its depth L, its canonical associate (Proposition 4.3) and its reduced word over S_5, optionally with its image in $PSL(2, \mathbb{F}_q)$ for an agreed q. A word of depth L takes $\lceil \log_2(6 \times 5^{L-1}) \rceil$ bits: 15 bits at depth 6.

Any agent recovers the word from the integer quaternion alone (Theorem 7.2) and the quaternion from the word by shifts and adds (Proposition 7.3). Two agents need agree only on S_5 and, to meet, on q. For integers rather than lattice points, the Einstein Tile's Form 2 carries three 10-bit residues modulo the prime triplet (881, 883, 887), decoded in O(1) by Garner's algorithm with closed-form inverses on [0, 690,017,701) (Fifteen Lemmas, Lemma 9 Part 2) **[PROVEN]**. This removes the interoperability barrier of Section 7.3: positions are compared as exact words, not as coordinates in a frame neither agent can see. For other primes $p \equiv 1 \pmod 4$ the same construction gives p + 1 generators and a (p + 1)-regular tree; p = 5 is the case whose generators are the cubic directions.

Editorial Note. Prime triplets (p, p + 2, p + 6) enter the standard only as CRT moduli: every triplet with p > 5 lies in one of two classes mod 30, (11, 13, 17) or (17, 19, 23), and the three moduli are pairwise coprime for every prime p > 3 (Fifteen Lemmas, Lemma 9 Parts 1 and 2). The generators that provably produce free, uniquely factorable hyperbolic structure are quaternions of prime norm.

8 The Sieve on the SHD-CCP 64-Bit Packet

The sieve runs on the Einstein Tile of the Fifteen Lemmas: it reads the Form 1 Hurwitz core in exact 64-bit arithmetic, certifies prime-norm anchor packets with a 2,017-byte table, gives every lattice address a frame-free canonical form, and prunes a stream losslessly where the TQF, read correctly, allows. Anchored to the realized cubit, its Triple Quad Formula stays exact in 64-bit integers.

Status: [DEFINITION], [PROVEN] and [COMPUTED]. Sources: Fifteen Lemmas, §6.1 (the Einstein Tile), Lemma 5 Parts 5 and 6(a), Lemma 4 Part 2 and Lemma 15; BioChain AI (2026e) for the decoding stages; Hoard (2026a), §IV, for exact TQF arithmetic, most accurate method; Hoard (2026d), §§3.2 and 3.5, Proposition 3.1 and Remark 4.3, for the physical scale and the precision horizon.

8.1 The Einstein Tile

The Fifteen Lemmas fix the layout of the 64-bit SHD-CCP packet as the Einstein Tile, whose design rule is that a receiver knows how to decode a packet before decoding it: fixed fields at fixed positions, and a 4-bit Form ID that selects the reading of the rest (Fifteen Lemmas, §6.1). This paper adopts it; the companion maps the three layouts of the earlier companion papers onto it (Hoard 2026h, Section 4.6). Table 12 lists its fields and what the sieve does with each.

Table 12: The Einstein Tile (Fifteen Lemmas, §6.1) and the sieve's use of each field.

Bits	Field	Content	Use by the sieve
61–63	Cycle step k	The exact spinor state (1 + i)^{k+1}	Carries the spinor sign that Form 1 normalizes away
58–60	Wheel unit	One of the 8 units mod 30, split into B_+ and B_- by the Legendre symbol mod 5	Proposed: the class of N mod 30 for an anchor packet (Section 3.1; Proposition 3.3)
56–57	Structure selector	Which structure, or the implied fourth	Not used
56–63	South Halo header	Bits 56–63 read as one byte	A cache filter only, never an integrity check (Fifteen Lemmas, Lemma 15)
48–55	Gate byte	Locus, C_9 or D_3 sector and sense, harmonic flag	Not used
44–47	Step-multiplexed index	Circle phase, half-cube, ghost index or frame coordinate	Not used
40–43	Form ID	Selects the reading of the core word and gate byte	The sieve acts on Form 1; Form 2 carries CRT addresses
36–39	Stream slot	Engine position or projection	Not used
32–35	Scale register	Dyadic level, Fibonacci index or frequency	Scale in dyadic multiples 2^k C_0 of the cubit $C_0 = \lambda/16$ (Hoard 2026d, §3.2)
0–31	Core word	Form 1: doubled Hurwitz coordinates 2Q, gcd-reduced and sign-normalized, with a loss flag in the top bit of the w byte	The sieve's input

Definition 8.1 (The Sieve's Reading of the Tile). [DEFINITION]

The sieve reads a packet only when its Form ID is 1 and the loss flag is clear, so that the core is an exact Hurwitz quaternion Q with doubled coordinates D = 2Q, $D_w \in [0, 127]$ and $D_x, D_y, D_z \in [-127, 127]$, all of one parity. Its outputs (the norm, the wheel class, the certificate and the canonical address) are derived quantities that every receiver recomputes from the core; none of them is stored in the core.

Remark (Physical Scale). [DEFINITION] With the scale register in cubits, the Form 1 lattice has a physical reading. At level 0 adjacent nodes are $C_0 = \lambda/16$ apart, a 4 × 4 × 4 block is the voxel cube of the cubit installment, whose edge is λ/4, and one node step along a plane wave advances the phase by π/8 (Hoard 2026d, Proposition 3.1, **[PROVEN]** there). Level k multiplies every length by 2^k, so level 4 makes the node spacing one wavelength. The reading adds no

constraint on the sieve, whose arithmetic is unit-free; it fixes what a lattice address means when two agents exchange physical positions (Section 9.2).

8.2 The Sieve Pipeline on One Packet

Each step below is exact; steps 2 to 6 store nothing.

1. Check that the Form ID is 1 and the loss flag is clear; otherwise the core is not exact and the sieve does not run.
2. Compute N(Q) = (D_w² + D_x² + D_y² + D_z²)/4 ≤ 16,129, a 14-bit integer.
3. Apply the wheel: keep N if N mod 30 is a unit or N ∈ {2, 3, 5} (Proposition 3.3). 34.26 % of encodable cores pass.
4. Certify with one lookup in a 2,017-byte bitmap of the 1,877 primes up to 16,129. Cores of prime norm, 14.36 % of the 265,314,240 encodable states, are anchor packets: irreducible and primitive (Proposition 4.1). They reach 1,712 of the 1,877 primes; the others need coordinates outside the byte range.
5. For an address, take a copy of Q to its canonical associate (Proposition 4.3). The stored core keeps the tile's sign normalization, because it encodes a rotation.
6. Address the core: when N = 5^L, factor it into S_5 (Theorem 7.2), which fits the Form 1 range to depth 5 (3,750 addresses at that depth); otherwise factor it by prime norms in the agreed order (Theorem 4.4). Integers in [0, 690,017,701) can instead travel as Form 2 CRT addresses (Section 7.7).
7. Key the Codex on a hash of the canonical form, not of the raw bits, so that one shape seen in two frames is one Codex entry (BioChain AI 2026e, Stage 4). The key is for lookup; integrity comes from the Merkle root of Section 9.6.
8. Across a stream of packets, prune a packet whose core is the linear prediction of its neighbours (Proposition 8.3) and store the run length.

Proposition 8.2 (Packet-Scale Arithmetic Is Exact in 64 Bits). **[PROVEN]** Uses Fifteen Lemmas, Lemma 5 Part 6(a).

For three Form 1 cores with difference vectors u and v in doubled units, the quadrea is

$$A_R = S^2 - 2\left(Q_1^2 + Q_2^2 + Q_3^2\right) = 4\left(Q_2 Q_3 - \langle \mathbf{u}, \mathbf{v} \rangle^2\right) \geq 0,$$

with Q_2, $Q_3 \leq 258{,}064 < 2^{18}$ and $Q_2 Q_3 < 2^{36}$. Both forms are exact in 64-bit integers; the Gram form needs 36 bits, the S^2 form 40.

Proof. With A at the origin, $Q_2 = |u|^2$, $Q_3 = |v|^2$ and $Q_1 = |u - v|^2$, and expanding gives the Gram identity, which is non-negative by Cauchy–Schwarz. Doubled coordinates differ by at most 254 per axis, so each quadrance is at most $4 \times 254^2 = 258{,}064$ and $Q_2 Q_3$ is at most $66{,}597{,}028{,}096 < 2^{36}$; $S \leq 774{,}192$ gives $S^2 < 2^{40}$. ■

The Gram form also makes the collinearity test a Cauchy–Schwarz equality, $\langle u, v \rangle^2 = Q_2 Q_3$, and with the Weitzenböck bound $A_R \leq S^2/3$ (Fifteen Lemmas, Lemma 4 Part 2) it bounds the quadrea by the larger route.

Table 13: Arithmetic widths on the Form 1 core, at the cubit realization horizon and at the precision horizon of the source note.

Quantity	Form 1 core	Cubit realization, $Q \approx 2.3 \times 10^7$	Source-note horizon, $Q \approx 10^{12}$
Norm	14 bits	25 bits	40 bits
Quadrance of a difference	18 bits, doubled units	27 bits	40 bits
Quadrea, Gram form Q_2Q_3	36 bits: exact in int64	53 bits: exact in int64	About 80 bits: two widening 64 × 64 → 128-bit products (Fifteen Lemmas, Lemma 5 Part 6(a))
Quadrea, S^2 form	40 bits: exact in int64	57 bits: exact in int64	85 bits: needs 128-bit integers
Primality certificate	2,017-byte bitmap lookup	Miller–Rabin, 64-bit products	Miller–Rabin, 80-bit products
Generator step over S_5	Shift and add	Shift and add	Shift and add

Wide integers are needed only at the source note's horizon. On the Form 1 core the Gram form fits in 64 bits with 28 bits to spare, and on lattices anchored to the realized cubit with 11 to spare.

Remark (The Precision Horizon). **[HEURISTIC / COMPUTED]** The source note's angular resolution $\delta\theta \approx 1/\sqrt{Q}$ gives $Q \geq 10^{12}$ for six decimal places, coordinates near 10^6 and 40 of 64 bits (62.5 %). About 4.2×10^{18} points of $\mathbb{Z}^3$ and 4.9×10^{24} of $\mathbb{Z}^4$ lie below that horizon, against 37,607,912,018 primes. For lattices anchored to physical positions the cubit sets a lower ceiling. C_0 is realized with a relative standard uncertainty of 2.09×10^{-4}, dominated by the real-air equation of state (Hoard 2026d, §3.5), so angular detail finer than that carries no information, and the same rule gives $Q \approx 1/(2.09 \times 10^{-4})^2 \approx 2.3 \times 10^7$: coordinates up to 4,785, or 13 bits per axis. That is 44,000 times below the source note's horizon, and the Gram form there needs at most 53 bits (Table 13; Appendix A, section Z). The resolution rule $\delta\theta \approx 1/\sqrt{Q}$ is the heuristic part; the widths that follow from it are computed.

Remark (Why the TQF Is Evaluated Exactly). **[COMPUTED]** The cubit installment shows that double precision cannot evaluate the quadrea of a thin astrophysical triangle (Hoard 2026d, Remark 4.3). On the lattice the same failure decides collinearity wrongly. For A = (0, 0, 0), B = (10^6, 1, 0) and C = ($2 \times 10^6 + 1$, 2, 0), whose quadrances are near 4×10^{12}, the exact quadrea is 4 and the Gram form returns 4, while double precision returns 0 and declares the triangle collinear (Appendix A, section Z). The collinearity tests of Section 8.3 are therefore always evaluated in integers.

8.3 TQF Pruning, Restated

The decoding framework scans the 6 faces of the 4 × 4 × 4 cube, 3 layers deep, 18 scans in all, and prunes the middle node of any triple that satisfies the TQF, "ensuring there is no data loss" (BioChain AI 2026e, Stage 3).

> **Proposition 8.3 (What TQF Pruning Removes).** **[PROVEN]** Restates BioChain AI (2026e), Stage 3.
>
> *(1) Every triple of points on one lattice scan line satisfies the TQF, so the test as stated prunes every interior node whatever the data. (2) For samples (k, f_k) at*

equally spaced positions, the TQF holds for three consecutive samples exactly when $f_{k-1} - 2f_k + f_{k+1} = 0$; pruning those nodes and storing the end values and run length is lossless. (3) The relation $Q_1 + Q_2 = Q_3$ drawn in the same paper is the right-angle condition of Pythagoras, not collinearity.

Proof. (1) The TQF holds exactly for collinear triples (Wildberger 2005), and a scan line is a line. (2) Three points $(k - 1, f_{k-1})$, (k, f_k), $(k + 1, f_{k+1})$ are collinear exactly when the two slopes agree, that is when the second difference vanishes; the pruned values are then recovered by linear interpolation. (3) $Q(A, C) + Q(B, C) = Q(A, B)$ holds exactly when the angle at C is right. ■

Appendix A, section O, checks (2) on 20,000 random triples and (3) on 336 lattice triangles. The midpoint test "$2Q_3$" of the same stage is the special case $Q(A_1, A_3) = 4Q(A_1, A_2)$ of a collinear, equally spaced triple.

The TQF also limits where prime traps can sit along a flow line.

Theorem 8.4 (No Collinear Prime Triple). **[PROVEN]** Formalizes source §4; a second, lattice proof is Fifteen Lemmas, Lemma 5 Part 5.

Let three points have integer pairwise quadrances Q_1, Q_2, Q_3 satisfying the TQF $(Q_1 + Q_2 + Q_3)^2 = 2(Q_1^2 + Q_2^2 + Q_3^2)$. If Q_1 and Q_2 are prime, Q_3 is not. In particular, no collinear triple of lattice points has three prime quadrances.

Proof. The TQF holds exactly when $\sqrt{Q_3} = |\sqrt{Q_1} \pm \sqrt{Q_2}|$, so $Q_3 = Q_1 + Q_2 \pm 2\sqrt{(Q_1Q_2)}$. Since Q_3 is an integer, Q_1Q_2 is a perfect square; for primes this forces $Q_1 = Q_2 = p$, and then Q_3 is 0 or $4p$, neither of them prime. ■

Lattice proof. If A, B, C lie on a lattice line, write $B - A = sv$ and $C - A = tv$ with v primitive of norm N and s, t distinct non-zero integers. The quadrances are s^2N, t^2N and $(s - t)^2N$. The first two are prime only if $s^2 = t^2 = 1$ and $N = p$ is prime; then $t = -s$ and $(s - t)^2N = 4p$. ■

Whether a single lattice line nevertheless meets infinitely many nodes of prime norm is a case of Bunyakovsky's conjecture (companion, Problem 2); the companion also records how the source note's example triple 2, 3, 5 is a right triangle rather than a collinear one (Hoard 2026h, Section 3).

9 A Universal Lattice for Interoperable Alignment and Governance

The sieve's mathematics gives agents that share no frame, medium, clock or history what they need to communicate exactly: one number system, invariants they can compare from any frame, canonical addresses, a finite space to meet in, and certificates that do not depend on trust. Its use for AI alignment and governance is a design proposal **[HEURISTIC]**; each layer's mathematics carries its own tag.

Status: [PROVEN], [ESTABLISHED] and [HEURISTIC]. Sources: BioChain AI (2026h) for the master quaternion and the coherence Φ; BioChain AI (2026f), §4, for Dual-Helix governance and Proof-of-Manifold; BioChain AI (2026a) for the separation of proposal from certification; BioChain AI (2026e), Stages 1 and 4, for the translation requirement and the Codex; BioChain

AI (2026i), §9, for scale-normalized quads; Hoard (2026d), §§3.2, 3.5 and 4.1, Lemma 3.2 and Appendix E, for the unit, the clock windows and the cube group; Hoard (2026a) for the register; Butt et al. (2026); Horn (1987); Fifteen Lemmas, Lemmas 5, 7, 9 and 15, Proposition 11.4 and Theorems 7.5 and 7.10.

9.1 The Problem: Agents Without a Shared Frame

The decoding framework states the goal as "the minimum viable translation requirements for any higher-order complexity system to be able to communicate its implied logic through a physics and mathematically verifiable system" (BioChain AI 2026e, Stage 1). Two agents meeting for the first time may differ in five ways: the orientation of their frames, their unit of length, their clock, their procedural history, and the kind of data they hold. Data of different kinds become comparable only after each agent maps them into the lattice with its own encoder. The standard can then align the lattice representations exactly; whether two encoders mean the same thing by the same point is a semantic question that no geometry settles, and Section 9.6 keeps it outside every certificate.

9.2 The Layers of the Standard

Each layer of Table 14 removes one way in which two agents can differ, and each rests on a result of this paper or of the series.

Table 14: Layers of the universal lattice standard.

Layer	What the agents agree on	Supplied by	Status
1 Number	Exact integer and Hurwitz arithmetic; no floating-point state	Sections 4 and 6	**[PROVEN]**
2 Packet	The Einstein Tile and its Form IDs	Section 8; Fifteen Lemmas, §6.1	**[DEFINITION]**
3 Scale	One unit of length, the cubit $C_0 = v/(16 f_{cal})$, realized to 0.22 mm (k = 2), in the scale register	Hoard 2026d, §§3.2 and 3.5; Section 8.1	**[DEFINITION]**; realization **[COMPUTED]**
4 Tick	The 60-pulse register and the shared windows of integer carriers: 2 pulses for caesium alone, 30 pulses (0.5 s) for caesium, 40 Hz and 432 Hz together	Hoard 2026a, §V; Hoard 2026d, §4.1; Fifteen Lemmas, Proposition 11.4	**[DEFINITION]**; window arithmetic **[PROVEN]**
5 Frame	Canonical associates for addresses, sign normalization for rotations; invariants N, Q, A_R and Φ_M	Sections 4.2 and 9.3	**[PROVEN]**
6 Address	Reduced words over S_5; prime-norm factor sequences; Form 2 CRT residues	Sections 4.2 and 7	**[ESTABLISHED]**, **[PROVEN]**
7 Meeting space	The Ramanujan graph X(5, q) for an agreed prime q	Section 7.4	**[ESTABLISHED]**
8 Certification	Exact checks of every claim that crosses between agents, recorded under a 256-bit Merkle root	Section 3; BioChain AI 2026a; Fifteen Lemmas, Lemma 15	**[PROVEN]**
9 Governance	Proposals scored by geometry, accepted only on certificates	BioChain AI 2026f, 2026h	**[HEURISTIC]**

9.3 Non-Local Alignment by Invariants

Non-local alignment compares agents that share no neighbourhood and no frame, so it can use only quantities every frame computes alike.

> **Proposition 9.1 (Frame-Free Invariants and the Corrected Coherence).** **[PROVEN]** Gives the coherence of BioChain AI (2026h) its sign-invariant form.
>
> *Let u and v be unit quaternions, and let a frame change act on every voxel state by q ↦ uqv. (1) It preserves every norm, every pairwise quadrance, the TQF and the quadrea A_R. (2) The coherence Φ = (1/N)‖Σ q̂_i‖ of the voxel-alignment review is invariant under it, but not under q_i ↦ −q_i, which represents the same rotation. (3) The largest eigenvalue Φ_M of M = (1/N) Σ q̂_i q̂_iᵀ is invariant under both, lies between 1/4 and 1, and equals 1 exactly when all q̂_i agree up to sign.*
>
> *Proof. (1) The map is linear with |uqv| = |q|, so it is an isometry of $\mathbb{R}^4$, and every listed quantity is built from distances. (2) Σ uq̂_i v = u(Σ q̂_i)v has the same length; replacing one q̂_i by its negative changes the sum. (3) M becomes RMRᵀ for the orthogonal matrix R of the frame change, and q̂q̂ᵀ is unchanged by sign; M has trace 1, so its largest of four eigenvalues is at least 1/4, and it equals 1 exactly when M has rank one.* ■

Remark (Measured). **[COMPUTED]** For 64 voxel states near a common pole, Φ = 0.9969 before and after a random frame change and 0.0331 after random sign flips; Φ_M = 0.9938 in all three cases. Matching anchors between two frames by their sorted quadrances to all other anchors, a frame-free signature, is correct in 499 of 500 trials (Appendix A, sections S and T).

The sign correction is not cosmetic. At the Engine's two handshake windows, steps 2 and 6 of the cycle, the spinors are i and −i: the same half-turn with opposite spin sign, so a partner sees the same spatial orientation, and the spin sign is the one bit the geometry never exposes (Fifteen Lemmas, Theorem 7.5). A coherence score that changes when q becomes −q measures that hidden bit rather than alignment; Φ_M does not.

Scale-normalized quads, as in the folded-packet paper, make the signatures invariant under a change of unit as well (BioChain AI 2026i, §9), so two agents can match anchors before they agree on a length. The cubit supplies the physical counterpart.

Remark (Medium-Free Invariants). **[PROVEN]** Hoard (2026d), Lemma 3.2. *If an enclosure measures N_x, N_y and N_z cubits, each realized in the air it contains, its mode frequencies are 8 f_cal √(l²/N_x² + m²/N_y² + n²/N_z²), whatever the temperature, pressure and composition of that air.* Frame-free invariants remove the orientation an agent cannot share; cubit units remove the medium. Two agents that exchange lengths as cubit counts rather than in metres agree on the phase geometry at f_cal of everything they describe, though their air differs (Appendix A, section Z).

9.4 Local Alignment and Exact Certification

Once anchors are matched, local alignment finds the rotation between the frames and then certifies it.

> **Proposition 9.2 (Horn's Quaternion Solution).** **[ESTABLISHED]** Horn (1987).

For matched point sets, the rotation that minimizes the sum of squared distances is given by the unit eigenvector of the largest eigenvalue of a symmetric 4 × 4 matrix built from their cross-covariance.

Remark (Two Agents, Measured). **[COMPUTED]** In 500 trials, agent B held agent A's 12 integer anchors under an unknown cube rotation, in shuffled order, with noise of standard deviation 0.05. Horn's rotation was within 3.2×10^{-3} of the truth; rounded, it was an exact lattice symmetry that mapped every anchor exactly in all 500 trials, including the one trial in which signature matching had swapped two anchors of identical signature. The certificate, not the matching, decides. For a rotation that is not a lattice symmetry the solution is accurate to about 10^{-3} but can be certified only to a tolerance: local alignment is exact up to the symmetry group of the lattice.

That group is exact: $x \mapsto 2\rho_g(x)$ maps the core cube onto the voxel cube exactly when ρ_g lies in the 24-element rotation group O, whose spinor lift 2O contains all 24 Hurwitz units (Fifteen Lemmas, Lemma 5 Part 2) **[PROVEN]**. It is also the group the Engine's own tier maps generate: Rot_x(π/2) and Rot_y(π/2), the alternating tier orientations of Hoard (2026a, §2.3), generate all 24 rotations of O, and the adjacent-tier rotation h of Proposition 10.1 is one of them (Hoard 2026d, Appendix E; Appendix A, section Z) **[PROVEN]**. For a rotation outside O, the rational Cayley map gives exact rational unit quaternions without square roots (Fifteen Lemmas, Lemma 9 Part 3), so an agent can snap Horn's answer to a nearby rational rotation and certify that rotation exactly; only the claim that it is the true rotation stays approximate.

9.5 One-Way Compatibility and Complementary Encodings

An external result on quantum codes gives a precise model of partial compatibility **[ESTABLISHED]**. Butt et al. (2026) pair the tetrahedral [[15, 1, 3]] colour code with its Hadamard-transformed twin. A transversal CNOT preserves the code space only from the tetrahedral code to its twin; the reverse direction needs a pieceable round-robin CZ gate with a mid-circuit measurement of the stabilizers that stay invariant throughout. The Eastin–Knill theorem forbids a universal transversal gate set in any single code.

Read as an analogy **[HEURISTIC]**: two agents whose lattice charts differ by a global transform can translate cheaply in a direction whose checks are invariants of that transform; the other direction needs checks interleaved during translation, on the invariant set of Section 9.3, which here consists of norms, canonical words and images mod q. No single chart makes every operation cheap for every agent, so the standard fixes the invariants, not one privileged frame.

9.6 Consensus, Governance and the Limits of a Coherence Score

On the meeting graph X(5, q) any diffusive consensus, such as gossip averaging, converges in logarithmic time (Proposition 7.5): 3.80 time units to 1 % on 12,180 nodes, against 62.15 on a cubic torus of the same size and degree (Table 10).

The chiral engine paper splits governance between a fluid Helix A, the BioStrata Open Plane, and a crystalline Helix B, the HyperSim Genesis Chain, with Proof-of-Manifold verification by the TQF (BioChain AI 2026f, §4). The Dual-Helix principle fixes what each side may do. A certificate can establish that a packet is well formed, that its claims are consistent across frames, and that it was derived by stated operations from stated inputs. It cannot establish that its content

is true or that a decision is good. Φ measures agreement among packets, so a coherent error scores as high as a coherent truth, and once Φ becomes a target agents can raise it without becoming more accurate. Φ therefore belongs to Helix A as an advisory score, not to Helix B as a "Truth Score" (BioChain AI 2026h). The token economics of BioChain AI (2026f), §4.3, are outside the scope of this paper.

Two results of the Fifteen Lemmas fix what Helix B stores. A record of nested state is certified only by a collision-resistant hash chain: with $\tau_d = H(G_d \| \tau_{d+1})$, equal 256-bit roots for different states exhibit a collision of H, while any 64-bit label of a 12-layer state with 2^{72} configurations has some value with at least 256 preimages, and a linear checksum such as CRC-32 can be steered to any target (Lemma 15) **[PROVEN]**. And four records closed by $H_1H_2H_3H_4 = 1$ survive the loss of any one, since each is the inverse product of the other three, but not of two (Lemma 7 Part 3(d)) **[PROVEN]**: closure buys erasure tolerance, not capacity. What an exchange preserves is likewise fixed: the shared geometry at the handshake windows, the address, the hyperbolic distance and the root hash, but never a chain's history, since products are not injective (Fifteen Lemmas, Theorem 7.10).

Proposal (The Alignment Handshake). **[HEURISTIC]** A protocol for two agents with no shared history; every step is exact or explicitly certified, and Figure 1 draws it.

1. Exchange invariants: norm spectra, counts of prime-norm anchor packets, images mod q. No frame is needed.
2. Match anchors by frame-free signatures (Proposition 9.1).
3. Solve the relative frame by Horn's quaternion and snap it to the lattice symmetry group when it lies there (Proposition 9.2).
4. Certify exactly; on failure, return to step 1 with more anchors.
5. Exchange addresses as reduced words over S_5 or Form 2 residues, at clock windows both carriers share (Hoard 2026d, §4.1; Fifteen Lemmas, Proposition 11.4), and meet in X(5, q) for consensus (Section 7).
6. Record certificates on Helix B under a 256-bit Merkle root; keep heuristic scores such as Φ_M advisory on Helix A.

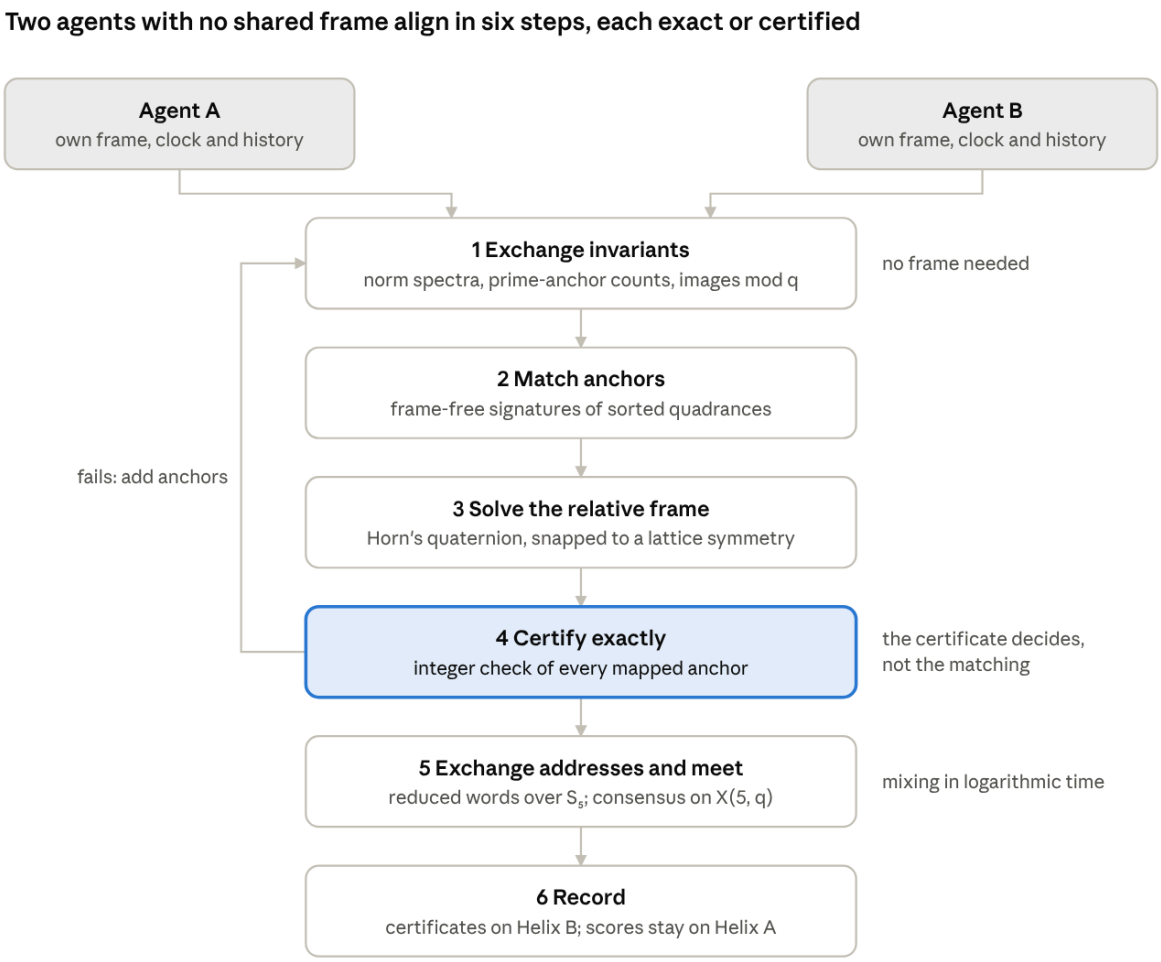


Figure 1: The alignment handshake. Steps 1 to 3 propose; step 4 certifies with integer arithmetic and loops back on failure; steps 5 and 6 share addresses and record the certificates.

Editorial Note. Nothing in this section claims that a geometric standard aligns the values of AI systems. It addresses the layer beneath: how systems that cannot see each other's frames can exchange states exactly, agree on what was said, and keep a record that does not depend on trusting either party. The companion fixes a preregistered test of the handshake between real systems, with its decision rule (Hoard 2026h, Problem 8).

10 Relation to the Geometric Engine of Time

The Geometric Engine of Time supplies the most accurate method at four points of the sieve: exact TQF arithmetic, the 30-node wheel, exact tier rotations and the closure clock; at three more it is the source of an identity that standard mathematics carries. The Fifteen Lemmas supply the proofs at each point, and the cubit installment sharpens three of them.

Table 15: Where the Geometric Engine of Time and its companion papers enter this paper.

Section	Series result	Role here	Most accurate method?
8, 9	Hoard 2026a, §IV (TQF area invariant), in exact arithmetic; Gram form from Fifteen Lemmas, Lemma 5 Part 6(a)	Collinearity, pruning, packet arithmetic	Yes. Exact in int64 at packet scale (Proposition 8.2); two 128-bit products at the horizon
3, 8	Hoard 2026a, §V, read as $\mathbb{Z}/30$ (Hoard 2026d, §4.6 and Theorem 4.9); Legendre split from Fifteen Lemmas, Lemma 4 Part 5	Wheel pre-filter before certification	Yes, for the packet. A proven bias of 3.75, the largest of any wheel fitting the 3-bit field (Propositions 3.2 and 3.3)
4, 7.5, 9.4	Hoard 2026a, §II.3 (alternating embedding); Fifteen Lemmas, Lemma 4 Part 4; Hoard 2026d, Appendix E	Tier rotation h as a Hurwitz unit; the tier maps generate the cube group O	Yes. Exact integer and half-integer arithmetic (Proposition 10.1)
9.2	Hoard 2026a, §V (30-node closure of 0.5 s); Hoard 2026d, §4.1; Fifteen Lemmas, Proposition 11.4	Clock windows of the handshake	Yes, for the closure arithmetic: 0.5 s is the shortest window shared by caesium, 40 Hz and 432 Hz
6, 7.5	Hoard 2026a, §I.1 ($W_\Sigma = 60$)	Exponent of 2I; order of PSL(2, 5)	Source only. A correspondence; 2I is not exactly representable (Theorem 6.1)
7.5	Hoard 2026a, §III.1 ($C_3 \times C_2$)	Abelian shadow of $PSL(2, \mathbb{Z}) = C_2 * C_3$	Source only. The modular group is the method
2	Hoard 2026b (cycle current); Fifteen Lemmas, Lemma 7 Part 3(e); Hoard 2026d, Proposition 4.8	Drift along TQF lines	Source only. Drift without a potential carries a current and produces entropy but leaves the stationary state uniform (Proposition 2.6)

Proposition 10.1 (The Engine's Tier Rotation Is a Hurwitz Unit). **[PROVEN]** The power table is Fifteen Lemmas, Lemma 4 Part 4; the cube group is Hoard (2026d), Appendix E.

The quaternion $h = (1 - i + j + k)/2$ that carries an even tier of the Engine to the next odd tier is one of the 24 Hurwitz units. Its powers are $h^2 = -\bar{h}$, $h^3 = -1$, $h^4 = -h$, $h^5 = \bar{h}$ and $h^6 = 1$, so h has order 6, and conjugation by h is the 120° rotation about $(-1, 1, 1)/\sqrt{3}$ that carries the tiers, an element of the cube group O that the tier orientations generate. Left multiplication by h preserves the norm of every Hurwitz integer and maps each prime shell onto itself.

Proof. All four components of h equal ±1/2, so h is a Hurwitz integer, $N(h) = 1$ and $Re(h) = 1/2$. A unit quaternion satisfies $q^2 - 2Re(q)q + 1 = 0$, so $h^2 = h - 1 = -\bar{h}$, since $\bar{h} = 1 - h$; the other powers follow, with $h^3 = -1$ the first real power and $h^6 = 1$ the first equal to 1. The conjugates $hi\bar{h} = -k$, $hj\bar{h} = -i$ and $hk\bar{h} = j$ give the rotation. Norms multiply, so $N(hq) = N(q)$. ■

Remark (Thirty). **[CORRESPONDENCE]** The reduced words of length 2 over S_5 number $6 \times 5 = 30$, the node count of the Engine's closure, and $720 = 24 \times 30$ is the number of Hurwitz units times that count. Both are equalities of numbers; no map between the structures is claimed.

Editorial Note. The Engine enters the sieve where exact arithmetic matters: its TQF, its wheel and its rotations are the most accurate methods available at those points, and the Fifteen Lemmas prove each. Its constants 60 and $C_3 \times C_2$ reappear in the hyperbolic

constructions of Section 7 as group-theoretic correspondences; the series should treat them as observations until a construction derives one from the other.

11 Conclusion

The Fokker-Planck prime sieve is the Helix A half of a Dual-Helix system: a geometric process that proposes where the primes are, paired with exact arithmetic that certifies them. Its mathematics is exact, and its use for communication between agents is stated as a design.

What the sieve is. On a lattice the sieve is a master equation. Its Gibbs state is positive on every node of a finite lattice, and on the infinite Hurwitz lattice exactly when the Hurwitz partition function converges; absorbing traps send all mass to the prime set with the harmonic measure; killing concentrates without selecting; and drift without a potential carries a current and produces entropy but leaves the stationary state uniform (Propositions 2.2 to 2.6). By Kolmogorov's criterion a potential selects at no stationary cost while a pump circulates without selecting. Certified traps are exact whatever the dynamics, and the Engine's wheel gives the proposer a proven bias of 3.75, the largest of any primorial wheel that fits the packet's 3-bit field (Propositions 3.1 to 3.3). The sieve lives in four dimensions, where every prime splits, every lattice point has a canonical associate and primitive points factor uniquely in a fixed norm order. The golden Clifford torus holds no lattice point, and its Fibonacci approximants carry the points q_m whose prime norms 7, 47 and 2207 are invisible to three dimensions (Theorems 5.2 and 5.3). Exact arithmetic has no drift, and exact delta schedules close within 12 repetitions or never (Section 6).

What it enables. The norm-5 quaternions $1 \pm 2i$, $1 \pm 2j$, $1 \pm 2k$ grow a hyperbolic tree over the cubic lattice by shifts and adds alone, with unique addresses and Ramanujan quotients on which diffusion mixes in logarithmic time, so procedurally generated cubic integers can be exchanged as exact words rather than as coordinates in a frame neither agent can see (Section 7). On the Einstein Tile the sieve runs in exact 64-bit arithmetic, certifies anchor packets with a 2,017-byte table, keys the Codex on canonical forms and prunes streams losslessly (Section 8). Anchored to the realized cubit, the lattice has a physical scale, a precision ceiling near $Q \approx 2.3 \times 10^7$ under which the Triple Quad Formula stays exact in 64-bit integers, and invariants that hold in any air. These define a universal lattice standard on which agents with no shared frame, medium or history align by invariants, solve their relative frame by Horn's quaternion, snap it to the cube group the Engine's own tier maps generate, and record every certificate under a Merkle root (Section 9).

Where the rest is. Every claim of the source note and of the companion papers that this paper corrects or replaces, the status of every claim, the open problems with their test protocols, and the relation to earlier versions of this installment are collected in the companion (Hoard 2026h).

A Computational Verification

One script, verify_fp_sieve.py, reproduces every number in this paper and in its companion (Hoard 2026h) and asserts every identity. Sections W to Y check every result imported from the

Fifteen Lemmas before it is used, and section Z every result taken from or strengthened by the cubit installment. It runs in about 90 seconds and passes all checks.

Running it. `python3 verify_fp_sieve.py` needs numpy, scipy and sympy, prints a report section by section, and writes the key values to fp_results.json.

Table A1: Sections of the verification script.

Section	What it checks	Outcome
A	Arithmetic of the source note	$Q \geq 10^{12}$; $2^{40} \approx 1.1 \times 10^{12}$; A_R(2, 3, 5) = 24
B	Theorem 8.4	No prime triple among prime pairs below 2,000; 10,227 collinear lattice triples checked
C	Theorem 4.2	4,511 of 17,984 primes below 200,000 are never three-square sums, all ≡ 7 mod 8
D	Prime nodes of the 4 × 4 × 4 and 4 × 4 × 4 × 4 lattices	25 and 94 nodes; 7 and 23 missing in 3D
E	Hurwitz integers; Proposition 10.1	24 units; norm counts 24σ_odd(n); no inert prime below 10,000; h of order 6
F	Proposition 2.2	Table 5; prime-node mass below 1 for every barrier
G	Proposition 2.3	Support on 25 prime nodes; λ_min = 1.402; mean absorption time 0.295
H	Integer width at $Q \approx 10^{12}$ (Table 13)	85 bits for S^2; int64 wraps
I	The wheel on integers	26.7 % of quadrances pass; every prime passes
J	Companion, Theorem 4.2	Exponents 12, 24 and 60; periodic schedules close; random 720-step products close in 92 of 2,000 runs
K	Proposition 7.6	SL(2, 3) and SL(2, 5) have the order profiles of 2T and 2I and are generated by S and T
L	Proposition 5.1; Theorem 5.2; companion, Section 4.1	r_1^2 and r_2^2 exact; one-sided delta off the torus in 1,000 of 1,000; two-sided exact in 1,000 of 1,000
M	Theorem 5.3	Table 8; even n ≤ 800 with L_n prime are 2, 4, 8, 16; L_{2^k} ≡ 7 mod 8; $L_{32} = 1087 \times 4481$
N	Companion, Proposition 4.1	Both trefoil symmetries to 4×10^{-15}
O	Proposition 8.3; Section 4.1	Scan-line triples collinear; second difference on 20,000 cases; right angle on 336 triangles; L(q) identities
P	Theorem 7.2; Proposition 7.3	6 × 5^{L−1} words, equal to direct enumeration for L ≤ 5; 2,000 of 2,000 factorizations; shift and add; Table 9
Q	Theorem 7.4; Proposition 7.5	Table 10; every nontrivial eigenvalue within 2√5
R	Byte-core statistics of an earlier packet reading	Kept for comparison with section Y (companion, Section 1)
S	Proposition 9.1	Φ frame-invariant, not sign-invariant; Φ_M invariant under both
T	Section 9.4	Signature matching 499 of 500; exact certificate 500 of 500
U	Proposition 3.2	β = 3.7500 on the integers 6 to 10^7
V	Proposition 2.4	Quasi-stationary prime-node mass 0.437 to 0.955, always below 1

Section	What it checks	Outcome
W	Fifteen Lemmas, Lemmas 4, 5, 9, 11, 13 and 14	Box primes 2, 3, 5, 7, 13; Gram form and Weitzenböck bound; Legendre split; triplet classes and Garner inverses for all primes to 20,000; finite rational orders 1, 2, 3, 4, 6; partition sums diverge at s = 2 and converge at s = 3; ν_Cs ≡ 30 mod 60
X	Fifteen Lemmas, Lemmas 7 and 8, Theorem 7.4, §7.2	Ghost map exact on 500 rational pairs; {4,3,5} constants; D_3 as six integer rotations; 13,824 unit chains give 24 products; uniform stationary state under line drift
Y	Section 8 on the Form 1 core	Norm ≤ 16,129; 1,712 of 1,877 primes reached; 14.36 % prime-norm states; 34.26 % pass the wheel; 2,017-byte bitmap; Gram form in 36 bits; S_5 words to depth 5
Z	Hoard 2026d, Propositions 4.8 and 4.6, Theorem 4.9, §§3.5 and 4.1, Lemma 3.2, Appendix E	Drift entropy production 0.0494 per step; Kolmogorov products equal for Gibbs rates; wheels mod 6, 30, 210 with biases 3, 3.75, 4.375 and fields of 1, 3, 6 bits; realization horizon Q = 2.29 × 10^7 with Gram form in 53 bits; double precision returns 0 for a quadrea of 4; closures of 2, 6, 10 and 30 pulses; tier rotations generate O, which contains h; cubit-scaled modes independent of v

B Notation and Construction Catalogue

For completeness, every symbol and construction this installment uses is listed below.

Table B1: Notation and construction catalogue.

Symbol / Object	Meaning
q = w + xi + yj + zk	Quaternion state
N(q) = qq̄	Norm, the sum of the squared components; the quadrance from the origin
Q, S, A_R	Pairwise quadrance; S = $Q_1 + Q_2 + Q_3$; quadrea A_R = $S^2 - 2\Sigma Q_i^2 = 4(Q_2Q_3 - \langle u, v\rangle^2)$ (Hoard 2026a, §IV; Fifteen Lemmas, Lemma 5)
Λ, ρ(t), V, D, β	Lattice nodes, probability vector, potential, diffusion constant, barrier parameter; rates k_ab = D e^(−β(V_b − V_a)/2), β = 1 unless stated
κ	Killing rate on composite nodes (Proposition 2.4)
L_TT, L_TA, λ_min	Generator on transient nodes, rates into traps, slowest transient decay rate
Z_H(β)	Hurwitz partition function 1 + Σ 24σ_odd(n)e^(−βV(n)) (Proposition 2.5)
w_k, ρ_k; σ (Section 2)	Forward weights and reverse valves of a drift on $(\mathbb{Z}/4)^3$; its entropy production per step (Proposition 2.6)
β (bias)	Oracle bias Pr_D(B)/Pr_U(B) (BioChain AI 2026a, Assumption 5.1); 3.75 for the wheel
B_+, B_-	Wheel units split by the Legendre symbol mod 5: {1, 11, 19, 29} and {7, 13, 17, 23}
Hurwitz integer; 2T	Quaternion with four integer or four half-integer components; its 24 units
2O, 2I	Binary octahedral (48) and icosahedral (120) groups; exponents 24 and 60; not exactly representable beyond 2T
L(q)	Matrix of left multiplication by q; $L(q)^T L(q) = N(q)I$
Canonical associate	The associate uq with odd positive real part, for addresses (Proposition 4.3)
φ; r_1, r_2	Golden ratio; radii of the golden Clifford torus, $r_1^2 = (5 + \sqrt{5})/10$
q(θ₁, θ₂)	r₁e^{iθ₁} + r₂e^{iθ₂}j, a point of a Clifford torus
F_m, L_m; q_m	Fibonacci and Lucas numbers; q_m = (F_m, F_{m+1}, F_{m−1}, F_m)
S_5	The norm-5 generators 1 ± 2i, 1 ± 2j, 1 ± 2k (Definition 7.1)

Symbol / Object	Meaning
σ (Section 7)	Shadow map from words over S_5 to $\mathbb{Z}^3$
X(5, q)	Lubotzky–Phillips–Sarnak Ramanujan graph on PSL(2, $\mathbb{F}$_q) or PGL(2, $\mathbb{F}$_q)
γ, t_mix	Spectral gap; time to 1 % total variation from a point start
{4,3,5}	The compact hyperbolic honeycomb of cubes, five around each edge
S, T	Generators of SL(2, $\mathbb{Z}$): S of order 4, ST of order 6
q̃ = (1/w, v/w)	Lorentz lift of a unit quaternion q = w + v (Proposition 7.7; Hoard 2026c)
C_rat, G	Rational Cayley map u ↦ (1 − ‖u‖², 2u)/(1 + ‖u‖²) and ghost map G = q̃ ∘ C_rat (Fifteen Lemmas, Lemma 8)
Q_h	Rational hyperbolic quadrance ⟨p̃, q̃⟩² − 1 = sinh² of the hyperbolic distance
Φ, Φ_M	Coherence of BioChain AI (2026h); its sign-invariant eigenvalue form (Proposition 9.1)
h = (1 − i + j + k)/2	The Engine's adjacent-tier rotation (Proposition 10.1)
30-node wheel	The 8 unit classes mod 30 (Hoard 2026d, §4.6); the best wheel for a 3-bit field (Proposition 3.3)
C_0, f_cal, u_rel	The cubit v/(16 f_cal), its calibration frequency and its relative standard uncertainty 2.09 × 10^{-4} (Hoard 2026d)
Einstein Tile; Form 1; D = 2Q	The 64-bit packet of the Fifteen Lemmas; its Hurwitz form; doubled core coordinates
τ_d	Merkle chain τ_d = H(G_d ‖ τ_{d+1}) (Fifteen Lemmas, Lemma 15)
Helix A, Helix B	Heuristic proposer and exact certifier (BioChain AI 2026a)

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Prime count. $\pi(10^{12}) = 37{,}607{,}912{,}018$ is the established value (OEIS A006880).

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